

Parametric Evaluation of Hybrid Single-Phase Cooling Strategies for High-Heat-Flux
Electronics

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B.S. in Mechanical Engineering, May 2025, The George Washington University

A Thesis submitted to

The Faculty of
The School of Engineering and Applied Science
of The George Washington University
in partial fulfillment of the requirements
for the degree of Master of Science

August 31, 2026

Thesis directed by

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Acknowledgments

I express my sincere gratitude to my advisors, Dr. Saniya LeBlanc and Dr. Cole Davis, for their guidance, expertise, and support throughout the development of this thesis. The feedback received throughout this process has challenged me to think critically both about the technical details of this work and more broadly as an engineer. I have learned a great deal about what it means to assume such an effort and what it takes to complete it, and could not have done so without their encouragement and insight. I would also like to thank Dr. Kartik Bulusu for chairing my defense committee and participating in the final feedback process.

I am also grateful to the faculty and staff of the Department of Mechanical and Aerospace Engineering at the George Washington University for the education, resources, and opportunities that have supported both my undergraduate and graduate studies. The knowledge and engineering perspective I developed during my time at GW provided the foundation necessary to undertake this research.

Abstract of Thesis

Parametric Evaluation of Hybrid Single-Phase Cooling Strategies for High-Heat-Flux Electronics

Modern high-performance electronic systems increasingly rely on three-dimensional (3D) integration, chiplet-based packaging, and high-power artificial-intelligence accelerators, which concentrate heat generation within smaller volumes while lengthening the thermal path from interior active layers to conventional package-level cooling surfaces. These trends create localized hotspots and thermal resistances that challenge remote heat sinks and cold plates. This thesis develops and evaluates a three-dimensional conjugate heat-transfer model of a hybrid single-phase cooling architecture that combines embedded microchannels with a distributed jet-impingement array. Water is used as the baseline coolant, and the model is intentionally operated at an extreme 700 W stacked-chip load to investigate the thermal-hydraulic limits of near-junction liquid cooling.

A parameterized COMSOL Multiphysics model is used to evaluate microchannel mass flow rate, channel count, channel open fraction, microchannel edge margin, and jet pitch ratio/array density. Increasing channel flow and channel count reduce peak temperature but impose increasingly severe pressure-drop penalties; reducing the uncooled edge margin suppresses perimeter hotspots; and increasing open fraction produces substantial initial benefit followed by diminishing thermal returns and reduced solid wall thickness. A variable-count, fixed-total-flow jet study identifies JPR 6 as the minimum-temperature configuration, while JPR 4 provides the better balance of peak temperature, uniformity, and pressure loss. The selected configuration uses 27 microchannels per cooling layer at an open fraction of 0.80, 25 g/s microchannel flow, a 0.10 mm edge margin, and a 4×4 array of $300 \mu\text{m}$ jets supplied by 4 g/s total jet flow. The model predicts a maximum chip temperature of approximately 374 K (101 °C), a maximum thermal resistance of 0.115 K/W, and an

ideal total hydraulic power of 8.38 W. Relative to the baseline configuration, maximum temperature decreases by approximately 103 K and maximum thermal resistance decreases by approximately 56%.

The selected design approaches the 100 °C target but requires approximately 333 kPa of microchannel pressure drop, showing that hydraulic head becomes a first-order constraint once aggressive embedded cooling is used. The results demonstrate that hybrid jet-microchannel cooling should be designed as a coupled thermal, hydraulic, geometric, and packaging problem rather than as a temperature-minimization problem alone. The model provides a framework for future studies of transient hotspots, adaptive flow allocation, alternative coolants, interface reliability, and plenum-level flow distribution in next-generation 3D electronic packages.

Table of Contents

Acknowledgments	ii
Abstract of Thesis	iii
List of Figures	ix
List of Tables	xi
List of Symbols / Nomenclature	xiii
1 Introduction	1
2 Literature Review	4
2.1 Thermal Challenges in High-Heat-Flux Electronic Systems	4
2.2 Microchannel Heat Sinks	5
2.3 Jet Impingement Cooling	5
2.4 Hybrid Jet-Microchannel Architectures	6
2.5 Adaptive and Flow-Regulated Cooling	6
2.6 Synthesis and Research Gap	7
3 Computational Model and Methodology	9
3.1 Modeling Progression and Validation Hierarchy	9
3.1.1 Reduced-Order Through-Thickness Framework	9
3.1.2 Preliminary Cooling-Architecture Models	11

3.1.3	Intermediate H100-Inspired Package Model	15
3.1.4	Transition to the Final Three-Dimensional Hybrid Model	21
3.2	Final 3D Model Geometry and Heat Load	21
3.3	Governing Equations and Performance Metrics	23
3.4	Materials and Thermophysical Properties	25
3.5	Physics Interfaces and Flow Model	26
3.6	Boundary and Initial Conditions	27
3.7	Baseline and Selected Geometric Configurations	30
3.8	Parameter Studies	32
3.9	Mesh, Solver, and Numerical Implementation	34
3.10	Model Assumptions and Limitations	36
3.11	Analytical and Numerical Validation Methods	38
3.11.1	Mass and Energy Conservation	38
3.11.2	Microchannel Pressure-Drop Estimate	39
3.11.3	Representative One-Dimensional Thermal-Resistance Model	41
3.11.4	Mesh Sensitivity	43
3.11.5	Single-Phase Consistency Check	44
3.11.6	Comparison with Published Hybrid-Cooling Performance	44
4	Results and Discussion	45
4.1	Baseline Performance	45
4.2	Effect of Microchannel Mass Flow Rate	47
4.3	Effect of Channel Count	49

4.4	Effect of Microchannel Edge Margin	51
4.5	Effect of Channel Open Fraction	52
4.6	Effect of Jet Pitch Ratio and Array Density	55
4.7	Selected Hybrid Configuration	57
4.8	Selected-Configuration Validation	59
4.8.1	Mass and Energy Conservation	59
4.8.2	Single-Phase Coolant Consistency	60
4.8.3	Analytical Microchannel Pressure-Drop Comparison	61
4.8.4	Representative One-Dimensional Thermal-Resistance Model	62
4.8.5	Mesh Sensitivity	65
4.8.6	Flow-Model Sensitivity	66
4.9	Thermal-Hydraulic Design Implications	68
4.9.1	Design Rules Emerging From the Parametric Studies	69
4.10	Comparison With Published Literature	71
4.11	Validation Summary	72
5	Conclusions	75
6	Future Work	78
6.1	Manifold and Flow-Distribution Design	78
6.2	Transient and Adaptive Cooling	79
6.3	Package Interfaces and Thermomechanical Reliability	79
6.4	Alternative Working Fluids and Two-Phase Cooling	80
6.5	Expanded Geometric and Multi-Objective Optimization	81

6.6	Experimental Demonstration and Validation	81
6.7	Chip-to-System Thermal Co-Design	82
	References	83
A	Computational Model, Data, and Reproducibility Archive	85
B	Additional Figures	87

List of Figures

1.1	Representative 3D-integrated package showing the vertical heat-removal challenge and the motivation for embedded near-junction cooling, based on the packaging and thermal-management concepts discussed in [1].	2
2.1	Comparison of representative single-phase microfluidic cooling architectures considered in the literature review: (a) parallel microchannel heat sink, adapted from [12]; (b) characteristic jet-impingement flow regions, including the free jet, stagnation region, and wall jet, adapted from [13]; (c) hybrid microchannel-jet-impingement heat sink, adapted from [5]; (d) manifold dual-microchannel architecture employing distributed coolant delivery and shortened flow paths, adapted from [14]; and (e) self-adaptive microfluidic cooling architecture employing temperature-regulated motion of a heat-sink pillar array, adapted from [4]. Panels (a) and (c)-(e) are adapted from works distributed under the Creative Commons Attribution 4.0 license ; panel (b) is adapted from a work distributed under the Creative Commons Attribution 3.0 license	8
3.1	Reduced-order thermal-resistance framework for the stacked hybrid cooling architecture. Vertical solid conduction is represented by the central thermal path, while the embedded microchannel layers and upper jet impingement region act as distributed coolant sinks.	11
3.2	Preliminary single-chip cooling architectures used during model development: (a) externally enhanced convection at $A_{\text{ratio}} = 100$, (b) direct water-jet impingement at $U_{\text{in}} = 1.0$ m/s, and (c) embedded microchannel cooling at a total flow rate of 9 g/s. A common 293-375 K temperature scale is imposed across all three panels to permit direct visual comparison.	15
3.3	Final three-dimensional hybrid cooling model showing isometric view of the complete computational architecture and cutaway view showing the three heat-generating silicon tiers, two embedded microchannel cooling layers, and upper jet-impingement region. Principal geometric parameters and coolant-flow directions are indicated.	23
3.4	Normal mesh used for the selected three-dimensional configuration: (a) complete computational domain and representative refinement in (b) the jet/nozzle region and (c) the embedded microchannel region. The normal mesh contains 852,151 domain elements.	35

4.1	Representative fields for the baseline configuration: (a) chip temperature with the maximum-temperature location indicated, (b) cross-sectional temperature through the stacked structure, (c) coolant velocity magnitude, and (d) coolant gauge pressure. Temperature panels use a common 293-480 K scale that is also applied to the selected-configuration comparison in Fig. 4.15.	46
4.2	Chip temperature metrics versus microchannel mass flow rate.	47
4.3	Channel pressure drop versus microchannel mass flow rate.	48
4.4	Thermal-hydraulic tradeoff for the microchannel mass-flow sweep.	49
4.5	Chip temperature metrics versus microchannel count.	50
4.6	Channel pressure drop versus microchannel count.	50
4.7	Remaining interchannel wall thickness versus microchannel count.	51
4.8	Effect of microchannel edge margin on chip temperature and perimeter-hotspot formation: (a) temperature metrics across the edge-margin sweep, (b) representative active-tier temperature field at a 0.50 mm edge margin, and (c) the corresponding field at a 0.10 mm edge margin. Panels (b) and (c) use identical spatial and temperature scales.	52
4.9	Chip temperature metrics versus channel open fraction.	53
4.10	Channel pressure drop versus open fraction.	54
4.11	Remaining wall thickness versus open fraction.	54
4.12	Chip temperature metrics for the variable-count, fixed-total-flow JPR/array-density study.	56
4.13	Jet total-pressure loss versus jet pitch ratio.	56
4.14	Thermal-hydraulic tradeoff for the jet-array study.	57
4.15	Thermal and coolant-flow fields for the selected hybrid configuration: (a) chip temperature with the maximum-temperature location indicated, (b) cross-sectional temperature through the stacked structure, (c) coolant velocity magnitude, and (d) coolant gauge pressure. The temperature panels use the same 293-480 K scale as the baseline fields in Fig. 4.1.	59
B.1	Mesh-sensitivity results for the selected hybrid configuration.	87

List of Tables

2.1	Synthesis of cooling approaches motivating the present hybrid design. . . .	7
3.1	Preliminary conventional-cooling results as a function of effective heat-rejection area.	12
3.2	Preliminary direct-jet-impingement results.	13
3.3	Functional heat-source allocation in the intermediate H100-inspired package model.	17
3.4	External-cooling response of the intermediate package model.	17
3.5	Jet-velocity sweep for the intermediate package model.	18
3.6	Representative results from the intermediate H100-inspired package-level cooling models.	20
3.7	Principal thermophysical properties used in the final computational model. .	26
3.8	Principal boundary and initial conditions used in the selected configuration.	29
3.9	Baseline configuration.	30
3.10	Selected configuration.	31
3.11	Parameter studies and principal fixed conditions.	32
3.12	Mesh levels used for the selected-configuration mesh-sensitivity study. . . .	34
4.1	Numerical screening metrics for the baseline and selected configurations. .	46
4.2	Thermal and hydraulic results of the microchannel mass-flow sweep. . . .	47
4.3	Thermal, hydraulic, and geometric results of the channel-count sweep. . . .	49
4.4	Thermal, hydraulic, and geometric results of the channel open-fraction sweep.	53
4.5	Results of the variable-count, fixed-total-flow JPR/array-density sweep. . .	55

4.6	Comparison of the baseline and selected hybrid configurations.	58
4.7	Mesh-sensitivity results for the selected configuration.	65
4.8	Exploratory Laminar-SST flow-model sensitivity for the selected configuration.	67
4.9	Design rules derived from the investigated parameter ranges.	70
4.10	Summary of numerical validation and physical-consistency checks for the selected configuration.	72
A.1	Principal contents of the computational reproducibility archive.	86

List of Symbols / Nomenclature

A	Area
c_p	Specific heat at constant pressure
D_h	Hydraulic diameter
D_j	Jet nozzle diameter
H	Jet nozzle-to-surface standoff distance
h	Convective heat-transfer coefficient
k	Thermal conductivity
$\dot{m}$	Mass flow rate
N_{ch}	Number of microchannels
N_j	Number of jets
p	Static pressure
p_j	Jet center-to-center pitch
P_{hyd}	Ideal hydraulic power transferred to coolant
Q_{gen}	Total imposed heat generation
q'''	Volumetric heat generation
R_{th}	Thermal resistance
T	Temperature
ΔT_{chip}	Difference between maximum and minimum active-tier temperatures
$\mathbf{u}$	Fluid velocity vector
w_{ch}, h_{ch}	Rectangular microchannel width and height
ρ	Fluid density
μ	Dynamic viscosity
$\boldsymbol{\tau}$	Viscous stress tensor
A_{wet}	Wetted heat-transfer area

f	Darcy friction factor used in reduced-order correlations
Nu	Nusselt number
Pr	Prandtl number
Re	Reynolds number
$\dot{V}$	Volumetric flow rate
Q_{vd}	Volumetric heat generation due to viscous dissipation
p_{abs}	Absolute pressure
p_{ref}	Reference pressure
T_{bulk}	Bulk coolant temperature
T_{sat}	Saturation temperature
ΔT_{sub}	Local coolant subcooling margin
JPR	Jet pitch ratio, p_j/D_j
TSV	Through-silicon via
TIM	Thermal interface material
CHT	Conjugate heat transfer
HPC	High-performance computing
HBM	High-bandwidth memory
OF	Channel open fraction
RANS	Reynolds-averaged Navier-Stokes
SST	Shear Stress Transport

Chapter 1: Introduction

Three-dimensional integration, chiplet-based architectures, high-bandwidth-memory stacks, and heterogeneous packaging have become central strategies for extending electronic-system performance as conventional transistor scaling becomes more difficult and expensive. These approaches shorten electrical interconnects and increase bandwidth density, but they also concentrate power dissipation into compact packages and place active layers behind additional silicon, bonding, underfill, and interconnect interfaces. The result is a fundamental co-design problem: the same packaging technologies used to improve electrical performance can also make heat harder to remove [1]. This challenge is increasingly visible in high-performance computing and artificial-intelligence hardware, where the industry trajectory is toward greater compute density and more direct integration of cooling with the silicon and package [2], [6].

The thermal consequences are particularly severe in 3D systems. Interior heat-generating tiers have limited access to conventional external heat sinks, while localized hotspots can reach kilowatt-per-square-centimeter scales. Elevated junction temperature and large spatial thermal gradients reduce electrical efficiency, accelerate degradation of interfaces and interconnects, and constrain how densely future logic and memory can be integrated. Thermal management is therefore no longer a peripheral packaging task; it is increasingly a first-order design constraint governing attainable performance, reliability, and package architecture [1], [3].

The final 3D model in this thesis intentionally stresses this problem. A 700 W total heat load is distributed through three active tiers sharing a $5\text{ mm} \times 5\text{ mm}$ projected footprint. On a footprint-normalized basis, this corresponds to approximately 2.8 kW/cm^2 of aggregate stack power, placing the numerical study near the extreme end of demonstrated embedded-cooling research rather than at a conventional package operating point. Jet-enhanced manifold microchannels have recently demonstrated cooling at the $3,000\text{ W/cm}^2$

scale, illustrating both the relevance and severity of this operating envelope [10]. The purpose of the present loading is therefore not to reproduce a specific commercial processor, but to test whether a compact hybrid single-phase architecture can remain thermally effective as the package approaches the operational limits motivating embedded cooling.

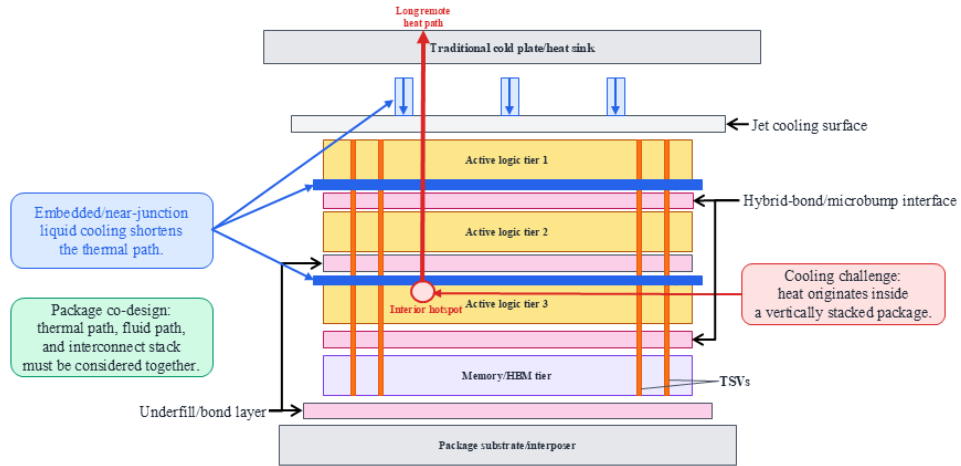


Figure 1.1: Representative 3D-integrated package showing the vertical heat-removal challenge and the motivation for embedded near-junction cooling, based on the packaging and thermal-management concepts discussed in [1].

Conventional forced-air heat sinks and remote liquid cold plates remain useful at moderate heat flux, but they require heat generated inside the package to conduct through multiple layers before reaching the coolant. Single-phase microfluidic cooling offers a more direct alternative. Microchannels provide large wetted area and short conduction paths for distributed heat removal, while jet impingement can generate very high local heat-transfer coefficients through stagnation-region boundary-layer disruption [3], [7]. Neither approach is ideal by itself: straight microchannels can impose high pressure drops and streamwise temperature rise, whereas jet arrays can suffer from spatial nonuniformity, crossflow interaction, and nozzle pressure losses. Hybrid jet-microchannel systems seek to combine the localized benefit of jets with the distributed bulk transport of channels [5], [9].

The research objective is therefore to determine how a hybrid single-phase jet-

microchannel architecture behaves when geometric coverage, coolant allocation, temperature uniformity, and hydraulic resistance are varied together. A sequence of reduced-order, preliminary single-level, intermediate package-level, and final stacked three-dimensional COMSOL models is used to progress from basic thermal-physics verification and architecture screening to a parameterized conjugate heat-transfer model of a stacked-chip cooling architecture. The final model evaluates microchannel mass flow rate, channel count, edge margin, channel open fraction, and jet pitch ratio/array density under a 700 W steady-state load. The design is assessed using maximum and average chip temperature, temperature span, thermal resistance, pressure drop, hydraulic power, numerical conservation, and geometric feasibility.

This thesis does not claim a global mathematical optimum or a complete commercial cooling loop. Instead, it identifies design trends and a selected configuration within the investigated design space, then evaluates the cost of that thermal performance in pressure head, pumping power, and remaining structural material. In this sense, the central contribution is a reproducible thermal-hydraulic modeling framework that translates the literature's qualitative argument for embedded hybrid cooling into quantitative design guidance for extreme 3D electronic packages.

Chapter 2: Literature Review

The literature converges on a common system-level problem: as electronic packaging moves heat generation closer together and farther from conventional cooling surfaces, the required thermal resistance decreases at the same time that pressure-drop, manufacturability, and reliability constraints become more severe. The following review therefore focuses not only on maximum heat-transfer capability, but on how microchannels, jets, hybrid architectures, and adaptive concepts redistribute the thermal and hydraulic burden within an advanced package.

2.1 Thermal Challenges in High-Heat-Flux Electronic Systems

Three-dimensional integration and heterogeneous packaging increase compute density by vertically and laterally integrating logic, memory, and specialized functions. The same geometry raises thermal resistance by adding material interfaces and by locating some heat sources away from the package exterior. Reviews of microfluidic 3D-IC cooling identify embedded liquid cooling as a particularly attractive response because coolant can be placed closer to the heat-generation sites rather than relying exclusively on heat spreading to a remote surface [1]. Recent industry demonstrations similarly emphasize direct-to-silicon and near-junction microfluidics for high-power AI hardware [2], [6].

Single-phase liquid cooling is attractive because it avoids the phase-instability and vapor-management complexity of boiling systems while providing much larger heat capacity and thermal conductivity than air. Its primary penalties are hydraulic: small passages and high velocities produce pressure loss, pumping demand, and flow-distribution challenges. These tradeoffs motivate the use of geometry and flow allocation as co-equal design variables rather than treating coolant flow as an unlimited resource [3].

2.2 Microchannel Heat Sinks

Microchannel heat sinks remove heat through a dense network of small passages located near the heat source. Their thermal advantage arises from high surface-area-to-volume ratio, short solid conduction distances, and high convective coefficients. The microchannel literature demonstrates strong dependence on hydraulic diameter, aspect ratio, channel count, manifold design, and flow rate [3].

Straight parallel channels also exhibit three recurring limitations. First, reducing hydraulic diameter or increasing channel length can rapidly increase pressure drop. Second, coolant warms and thermal boundary layers develop along the flow direction, producing streamwise temperature nonuniformity. Third, parallel networks can suffer from maldistribution when inlet and outlet manifolds do not distribute pressure evenly. Manifold microchannels, multilayer structures, interrupted passages, fins, and other enhancements seek to reduce these penalties, but generally increase geometric or fabrication complexity [3], [4].

2.3 Jet Impingement Cooling

Jet impingement directs coolant through one or more nozzles onto a heated surface. Near the stagnation point, momentum is redirected and the thermal boundary layer is repeatedly thinned, producing some of the highest local single-phase heat-transfer coefficients available. Jet performance depends strongly on nozzle diameter, jet Reynolds number, pitch, standoff distance, array arrangement, and crossflow environment [7], [9].

The principal limitation is spatial localization. A strong stagnation zone does not automatically produce uniform cooling across an entire die. Dense arrays increase surface coverage but can create jet interaction and crossflow, while sparse arrays place more flow through each nozzle but leave larger areas between stagnation zones. This tradeoff is especially important when total coolant flow is constrained, because jet count then changes

the momentum available to each nozzle.

2.4 Hybrid Jet-Microchannel Architectures

Hybrid cooling combines the distributed heat-removal area of microchannels with localized boundary-layer disruption from impinging jets. Xiao et al. experimentally demonstrated a hybrid microchannel and jet-impingement heat sink with a reported thermal resistance near 0.11 K/W, illustrating the potential of the combined mechanism [5]. More recent jet-enhanced manifold microchannels extend the concept toward extremely high heat-flux operation by shortening flow paths and distributing impingement throughout the cooling structure [10].

The literature suggests that the benefit of hybridization does not come merely from adding a second cooling mechanism. Jet placement, channel geometry, total flow allocation, and pressure loss must be co-designed. A jet array that minimizes local temperature may not minimize total pumping demand, and a channel geometry that maximizes wetted area may leave insufficient wall thickness or impose excessive hydraulic head. This thesis adopts that coupled interpretation as the basis of the parametric design studies.

2.5 Adaptive and Flow-Regulated Cooling

Advanced electronic workloads are spatially and temporally nonuniform, which means cooling systems designed only for the worst steady-state hotspot can waste pumping power during lower-load periods. Adaptive cooling concepts seek to redistribute coolant according to local thermal demand. Xiang et al. modeled a self-adaptive microfluidic architecture using temperature-regulated metal pillars, while Feng et al. investigated hydrogel-enhanced adaptive jet impingement using through-silicon-via concepts [4], [8]. Broader programmable-microfluidics research also demonstrates active and passive methods for changing local flow resistance and routing [11].

Adaptive flow control is not implemented in the final steady-state model developed in

this thesis. It remains an extension of the fixed hybrid architecture. The present study instead establishes which geometric and hydraulic parameters most strongly control temperature and pressure, providing a design basis for deciding where future adaptive control would be most useful.

2.6 Synthesis and Research Gap

Table 2.1 summarizes the role of each architecture. The literature indicates that microchannels are strong distributed coolers but can become hydraulically expensive; jets provide exceptional local cooling but require careful spacing and flow allocation; hybrid architectures can combine both strengths but create a multi-objective design problem; and adaptive systems offer a mechanism for responding to nonuniform loads but add control and fabrication complexity.

Table 2.1: Synthesis of cooling approaches motivating the present hybrid design.

Architecture	Primary advantage	Primary limitation / design implication
Microchannels	Distributed near-junction heat removal; large wetted area; short conduction paths.	Pressure drop, streamwise warming, flow maldistribution, and wall-thickness constraints.
Jet impingement	Very high local stagnation-region heat transfer; strong hotspot suppression.	Spatial nonuniformity, array interaction, crossflow, and nozzle pressure loss.
Hybrid jet-microchannel	Combines local boundary-layer disruption with distributed bulk coolant transport.	Requires co-optimization of jet density, channel geometry, total flow allocation, and hydraulic head.
Manifold / jet-enhanced channels	Shorter flow paths and improved coolant distribution; demonstrated extreme heat-flux capability.	Increased integration and fabrication complexity.
Adaptive cooling	Redistributes coolant toward changing hotspots and may reduce overcooling.	Added material/control complexity; transient response and reliability remain open questions.

The remaining gap is therefore not evidence that hybrid liquid cooling can transfer heat; that has already been demonstrated. The need is for modeling frameworks that expose the tradeoffs among hotspot suppression, thermal resistance, pressure drop, temperature

uniformity, and geometric feasibility for stacked-chip configurations. The present work addresses that gap using a staged COMSOL methodology and discrete parametric studies, with the final design treated as a selected engineering compromise rather than a single-variable temperature optimum.

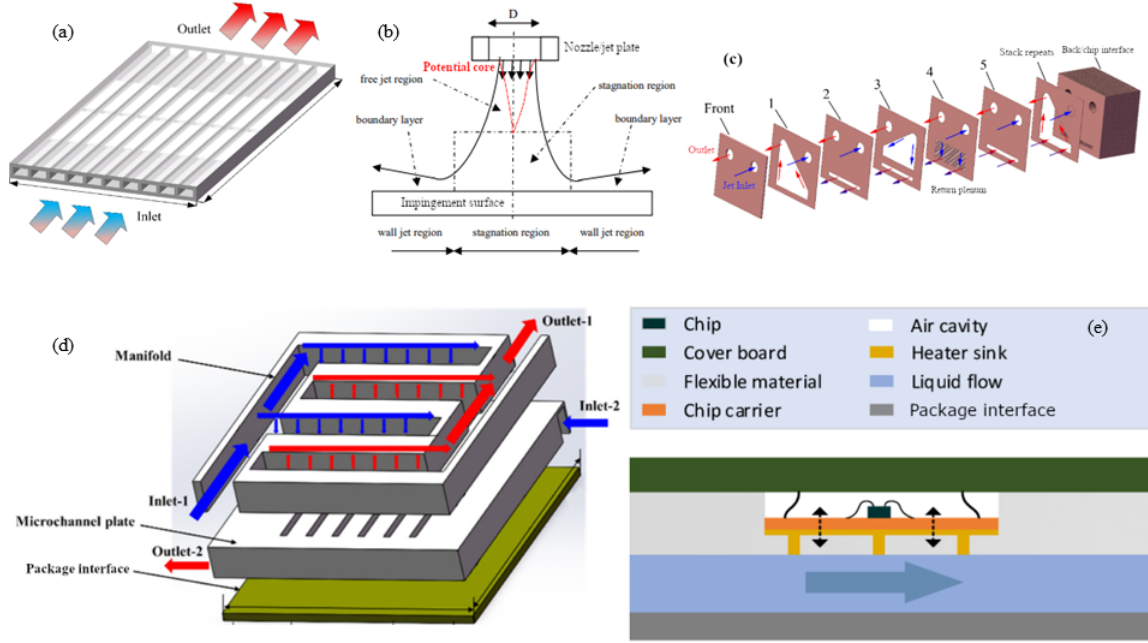


Figure 2.1: Comparison of representative single-phase microfluidic cooling architectures considered in the literature review: (a) parallel microchannel heat sink, adapted from [12]; (b) characteristic jet-impingement flow regions, including the free jet, stagnation region, and wall jet, adapted from [13]; (c) hybrid microchannel-jet-impingement heat sink, adapted from [5]; (d) manifold dual-microchannel architecture employing distributed coolant delivery and shortened flow paths, adapted from [14]; and (e) self-adaptive microfluidic cooling architecture employing temperature-regulated motion of a heat-sink pillar array, adapted from [4]. Panels (a) and (c)-(e) are adapted from works distributed under the [Creative Commons Attribution 4.0 license](#); panel (b) is adapted from a work distributed under the [Creative Commons Attribution 3.0 license](#).

Chapter 3: Computational Model and Methodology

The modeling strategy was developed progressively rather than beginning directly with the final three-dimensional hybrid geometry. Reduced-order and preliminary single-level models were first used to verify the expected direction of heat flow and the characteristic behavior of conventional external cooling, impinging jets, and microchannels. The final 3D model then combined these mechanisms in a parameterized conjugate heat-transfer framework. This progression reduced debugging ambiguity and provided physical context for interpreting the full model.

3.1 Modeling Progression and Validation Hierarchy

3.1.1 Reduced-Order Through-Thickness Framework

A reduced-order thermal-resistance framework was used to interpret the dominant through-thickness heat-transfer mechanisms before and alongside development of the full three-dimensional conjugate model. The framework is not intended to reproduce local hotspots, lateral heat spreading, or the detailed flow distribution within the microchannels and jet chamber. Instead, it represents the stacked architecture as a one-dimensional thermal ladder in which heat conducts through the solid structure while being removed at multiple coolant interfaces.

The final architecture consists of three heat-generating silicon tiers separated by two copper microchannel cooling regions, with an additional jet-impingement cooling boundary above the upper silicon tier. A purely series resistance model would therefore be inappropriate because heat is not required to travel through the complete stack before reaching a coolant sink. Instead, the two embedded microchannel layers and the upper jet-cooled surface act as parallel heat-removal branches distributed along the primary through-thickness conduction path.

For a representative solid segment, the conduction resistance is

$$R_{\text{cond},i} = \frac{L_i}{k_i A_i}, \quad (3.1)$$

where L_i , k_i , and A_i are the characteristic conduction distance, thermal conductivity, and representative heat-transfer area, respectively. For a coolant interface, the corresponding convective resistance is represented by

$$R_{\text{conv},i} = \frac{1}{h_i A_{\text{wet},i}}. \quad (3.2)$$

Each embedded cooling level can therefore be represented approximately by a branch resistance

$$R_{\text{mc},i} = R_{\text{cond},i} + R_{\text{conv},\text{mc},i}, \quad (3.3)$$

while the upper cooling path is represented by

$$R_{\text{jet}} = R_{\text{cond},\text{top}} + R_{\text{conv},\text{jet}}. \quad (3.4)$$

The resulting network is analogous to a thermal ladder: vertical conduction through the stacked solid structure forms the principal through-thickness path, while the two microchannel layers and the jet-impingement surface provide distributed thermal sinks. This representation captures the principal architectural advantage of embedded cooling: heat can be removed near the tier in which it is generated rather than being conducted through the entire package to a single external cooling surface.

The reduced-order framework is used later as an independent thermal-scale check on the three-dimensional model. The analytical calculation employs representative conduction lengths and independently estimated convective heat-transfer coefficients; it is therefore interpreted as a simplified engineering estimate rather than an exact analytical solution of the stacked conjugate heat-transfer problem.

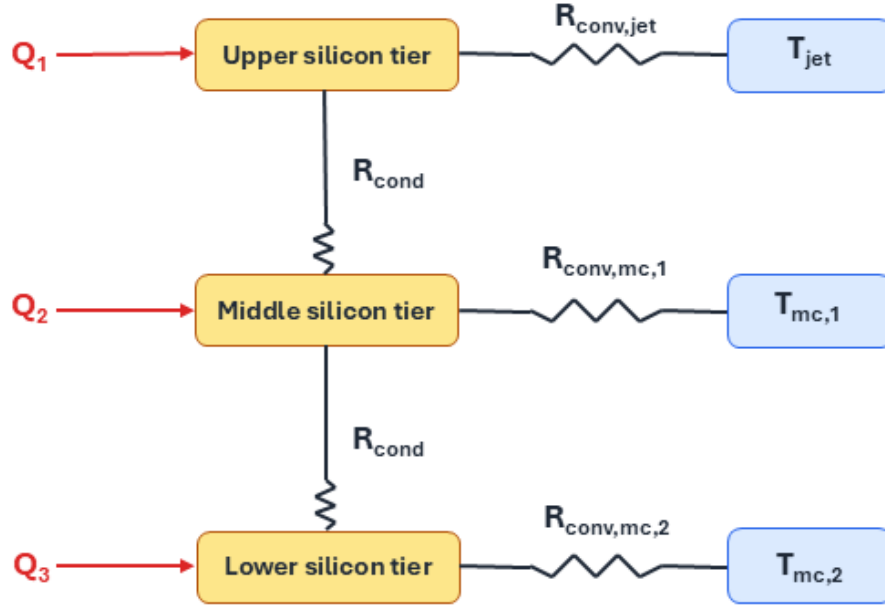


Figure 3.1: Reduced-order thermal-resistance framework for the stacked hybrid cooling architecture. Vertical solid conduction is represented by the central thermal path, while the embedded microchannel layers and upper jet impingement region act as distributed coolant sinks.

3.1.2 Preliminary Cooling-Architecture Models

Prior to development of the stacked three-dimensional cooling architecture, three preliminary single-chip models were constructed to evaluate the thermal behavior and practical limitations of conventional external cooling, direct jet impingement, and embedded microchannel cooling. Although these models employed three-dimensional COMSOL geometries, they contained only a single heat-generating chip level and are therefore distinguished from the later vertically stacked model. Their purpose was architecture development rather than direct quantitative validation of the final configuration.

The preliminary chip was divided into four independently heated regions to represent a spatially nonuniform power distribution. The regional heat loads were 38.93, 39.09, 48.18, and 23.80 W, corresponding to a total chip heat generation of 150 W. The silicon chip thickness was 0.5 mm, and an effective thermal-contact resistance of $2 \times 10^{-5} \text{ m}^2\text{K/W}$ was applied between the relevant solid regions. Silicon and copper were used for the principal

solid domains, while water was introduced in the liquid-cooled models.

Conventional External Cooling

The first model represented conventional package-level cooling through a copper heat-spreading region connected to an externally cooled surface. Rather than explicitly resolving individual heat-sink fins and the surrounding air flow, the enhanced external surface area was represented through an effective convection boundary condition,

$$h_{\text{eff}} = h_{\text{air}} A_{\text{ratio}}, \quad (3.5)$$

where $h_{\text{air}} = 80 \text{ W}/(\text{m}^2 \text{ K})$ and A_{ratio} represents the ratio of effective heat-rejection area to the modeled base area. A 2-mm copper heat spreader was used and the ambient temperature was approximately 293 K.

Table 3.1: Preliminary conventional-cooling results as a function of effective heat-rejection area.

A_{ratio}	$h_{\text{eff}} \text{ (W/m}^2\text{K)}$	$T_{\text{max}} \text{ (K)}$	$T_{\text{avg}} \text{ (K)}$	$\Delta T_{\text{chip}} \text{ (K)}$
10	800	562	546	29
30	2,400	422	407	28
50	4,000	394	379	28
100	8,000	372	357	27
200	16,000	360	346	26

Increasing effective heat-rejection area substantially reduced chip temperature, demonstrating that conventional external cooling remained thermally viable at the 150-W development load when sufficiently aggressive heat rejection was provided. However, an area ratio of approximately 100, corresponding to an effective convective coefficient of approximately $8000 \text{ W}/(\text{m}^2 \text{ K})$, was required before the maximum temperature decreased to approximately 372 K (99 °C). This result illustrates the increasing external surface-area and packaging requirements associated with maintaining conventional cooling as heat generation rises. For vertically integrated devices, heat generated in interior tiers must additionally be conducted outward before reaching such an external cooling surface.

Direct Jet Impingement

A second preliminary model investigated direct water-jet impingement on the heated structure. The model used a 2-mm-diameter jet, a nominal 15-mm fluid height, a coolant inlet temperature of 293.15 K, and a zero-gauge-pressure outlet. Jet inlet velocity was varied from 0.1 to 2.0 m/s to characterize the sensitivity of the cooling response to jet momentum.

Table 3.2: Preliminary direct-jet-impingement results.

U_{in} (m/s)	T_{max} (K)	T_{avg} (K)	T_{min} (K)	ΔT_{chip} (K)
0.1	493	483	474	19
0.2	431	422	413	18
0.5	382	373	366	16
1.0	362	353	346	16
2.0	350	341	334	16

Direct impingement produced a strong reduction in chip temperature as jet velocity increased. Increasing inlet velocity from 0.1 to 1.0 m/s reduced maximum temperature by more than 130 K, and the 1.0 and 2.0 m/s cases maintained the maximum chip temperature below 100 °C. Temperature span remained approximately 16-19 K throughout the sweep, indicating comparatively uniform cooling of the single exposed chip.

The principal limitation identified by this model was therefore architectural rather than thermal. Direct impingement is highly effective when coolant has unobstructed access to the heated surface, but equivalent access cannot be provided readily to every active tier within a vertically integrated stack. Jet impingement was consequently retained as a complementary cooling mechanism rather than the sole internal cooling strategy.

Embedded Microchannel Cooling

The third preliminary model investigated embedded parallel microchannels. Twenty water channels were distributed across a 13.5-mm active width with a total coolant mass flow rate of 9 g/s. Each channel was 100 μm high and approximately 598.5 μm wide, corresponding to an open fraction of 0.90, a pitch of approximately 668.5 μm , and an interchannel wall

thickness of 70 μm . A 0.10-mm edge margin was retained around the active channel field. Water entered at 293 K and discharged through a zero-gauge-pressure outlet.

The microchannel model predicted a maximum chip temperature of 314 K (40 °C), an average chip temperature of 307 K (34 °C), and a chip temperature span of 16 K. The corresponding channel pressure drop was approximately

$$\Delta P_{\text{mc}} = 125.6 \text{ kPa.} \quad (3.6)$$

The preliminary microchannel model therefore provided the strongest thermal performance of the three architecture-development cases, demonstrating the benefit of placing coolant passages directly adjacent to the heat-generating structure. This improvement was accompanied by a substantial hydraulic penalty together with additional fabrication, fluid-distribution, sealing, and integration requirements.

Implications for the Stacked Architecture

The preliminary studies established the progression from conventional external heat rejection toward cooling located increasingly close to the heat-generating regions. Conventional cooling remained feasible at the development-scale heat load only with increasingly aggressive effective surface area. Direct jet impingement demonstrated excellent thermal performance where coolant could directly access the heated surface, but its accessibility becomes limited for buried tiers in a three-dimensional package. Embedded microchannels overcame this accessibility limitation by moving distributed liquid cooling into the solid structure itself, although at the cost of greater hydraulic and fabrication complexity.

These observations motivated the final hybrid architecture. Embedded microchannels were adopted as the principal distributed heat-removal mechanism for the stacked structure, while jet impingement was retained as a complementary mechanism in regions where direct fluid access could be provided. The final three-dimensional model therefore does not represent a simple scaling of any one preliminary configuration, instead combining the

mechanisms identified as most useful during the architecture-development stage.

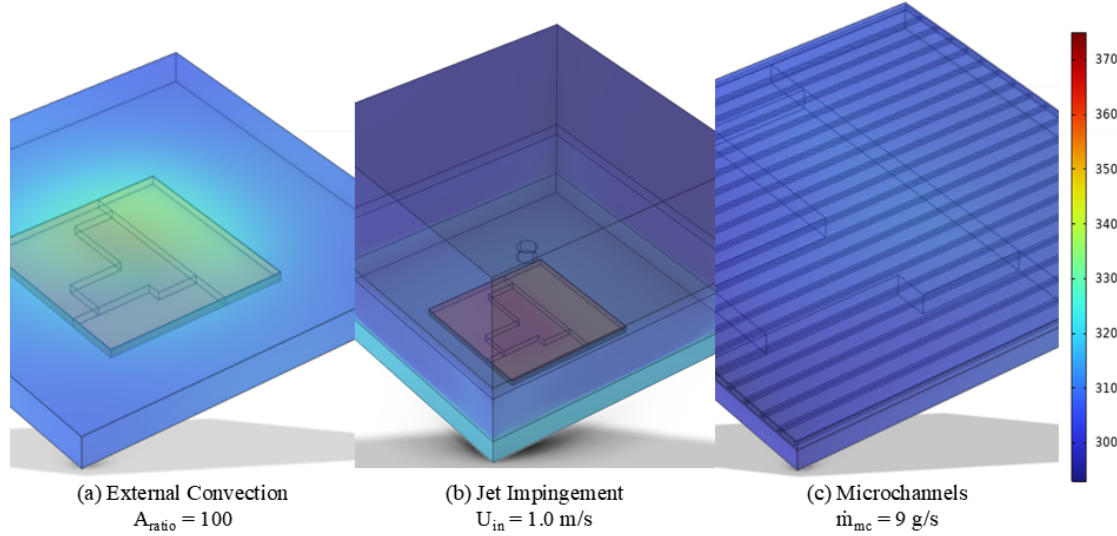


Figure 3.2: Preliminary single-chip cooling architectures used during model development: (a) externally enhanced convection at $A_{\text{ratio}} = 100$, (b) direct water-jet impingement at $U_{\text{in}} = 1.0 \text{ m/s}$, and (c) embedded microchannel cooling at a total flow rate of 9 g/s . A common 293-375 K temperature scale is imposed across all three panels to permit direct visual comparison.

3.1.3 Intermediate H100-Inspired Package Model

Following the preliminary single-chip architecture studies, an intermediate package-level model was developed to introduce a spatially heterogeneous heat-source distribution and substantially higher package power before construction of the final stacked-chip configuration. The model was inspired by the functional organization and power scale of the NVIDIA H100 Hopper accelerator rather than intended as a dimensionally exact reproduction of a commercial product.

The NVIDIA Hopper architecture contains streaming multiprocessors, cache and I/O subsystems, and high-bandwidth-memory regions, while the H100 SXM5 product is reported with a thermal design power of 700 W [15]. These published characteristics were used to establish the general package scale and functional organization of the intermediate model.

Individual regional power allocations were modeling assumptions introduced to create a heterogeneous thermal load and are not presented as manufacturer-reported component-level power data.

The COMSOL geometry used a nominal package footprint of approximately 52 mm $\times$ 33 mm. The heat-generating structure was divided into separate proxy regions representing streaming-multiprocessor, I/O, L2-cache, and HBM-related functions. Two streaming-multiprocessor regions were assigned 280 W each, two I/O regions were assigned 28 W each, and one L2-related region was assigned 84 W. These GPU-side regions therefore dissipated

$$Q_{\text{GPU}} = 2(280) + 2(28) + 84 = 700 \text{ W}. \quad (3.7)$$

Six HBM-related regions were additionally assigned 10 W each,

$$Q_{\text{HBM}} = 6(10) = 60 \text{ W}, \quad (3.8)$$

producing a total modeled package heat generation of

$$Q_{\text{package}} = 760 \text{ W}. \quad (3.9)$$

A direct COMSOL volume integration of the total heat-source term, $\text{ht} . Q_{\text{tot}}$, over the heat-generating proxy domains returned approximately 760 W, confirming that the implemented source assignments reproduced the intended 700 W GPU-side allocation plus 60 W of separately modeled HBM heat generation.

The resulting model is therefore interpreted as a severe H100-inspired package-level thermal proxy rather than a reconstruction of the internal power distribution of an NVIDIA product.

Table 3.3: Functional heat-source allocation in the intermediate H100-inspired package model.

Modeled region	Number	Power per region (W)	Total power (W)
Streaming-multiprocessor proxy	2	280	560
I/O proxy	2	28	56
L2-cache proxy	1	84	84
HBM proxy	6	10	60
Total package			760

External-Cooling Response

The package-level external-cooling model retained the effective surface-area formulation used during the preliminary architecture study. Increasing effective heat-rejection area reduced maximum temperature substantially, but the high package load produced temperatures well above the desired operating range for moderate area ratios.

Table 3.4: External-cooling response of the intermediate package model.

A_{ratio}	T_{max} (K)	T_{avg} (K)	T_{min} (K)	ΔT_{chip} (K)
10	1179	798	639	541
20	795	535	428	367
50	536	392	334	202
100	441	348	312	129
200	392	327	304	88

Even at an effective area ratio of 200, the maximum temperature remained approximately 392 K (118 °C). The result demonstrates that external heat rejection continues to provide substantial thermal benefit but becomes increasingly demanding in surface area as package power increases. This limitation becomes more restrictive in vertically integrated devices because heat generated in buried tiers must first be transported to an externally accessible cooling surface.

Jet-Impingement Response

A direct liquid-impingement model was also evaluated at inlet velocities of 1, 2, and 3 m/s. The resulting temperatures are summarized in Table 3.5.

Table 3.5: Jet-velocity sweep for the intermediate package model.

U_{imp} (m/s)	T_{max} (K)	T_{avg} (K)	T_{min} (K)	ΔT_{chip} (K)
1	401	372	363	37
2	365	338	329	36
3	352	327	318	34

Increasing impingement velocity from 1 to 2 m/s reduced maximum temperature by approximately 36 K, while increasing velocity from 2 to 3 m/s produced a smaller additional reduction of approximately 13 K. The 2 and 3 m/s cases reduced maximum temperature below 100 °C, confirming that direct liquid impingement remained thermally effective at the higher package power.

The principal constraint remained physical accessibility rather than heat-transfer capability. Surface impingement can provide strong local cooling where coolant access exists, but extending equivalent direct access to internal tiers becomes difficult in a vertically integrated architecture.

Embedded-Microchannel Response

The intermediate microchannel model used 16 parallel coolant channels extending across the approximately 52-mm package length. The converged geometry used a channel width of approximately 2.87 mm, a channel height of 0.40 mm, an interchannel wall thickness of approximately 0.34 mm, and an open fraction of 0.90. The channel array occupied an active width of approximately 51 mm.

A total water mass flow rate of 9 g/s was distributed across the 16 parallel channels. Integration of the inlet and outlet mass fluxes returned 9.00 g/s at both boundaries, confirming that the prescribed flow represented the total microchannel-branch flow rather than the flow

applied independently to each channel.

The resulting maximum chip temperature was approximately

$$T_{\max} = 381 \text{ K} = 107 \text{ }^{\circ}\text{C}. \quad (3.10)$$

The corresponding average and minimum chip temperatures were 323 K and 299 K, respectively, producing a chip temperature span of approximately

$$\Delta T_{\text{chip}} = 81 \text{ K}. \quad (3.11)$$

The microchannel pressure drop was approximately

$$\Delta P_{\text{mc}} = 883 \text{ Pa} = 0.883 \text{ kPa}. \quad (3.12)$$

The embedded microchannels therefore reduced the maximum temperature substantially relative to the package-level external-cooling cases, bringing the high-power package proxy to within approximately 7 °C of the 100 °C design objective at a total flow rate of only 9 g/s. The relatively low pressure drop reflects the comparatively large hydraulic dimensions of the intermediate channels and should not be interpreted as representative of the much smaller passages used in the final stacked configuration.

The comparatively large 81-K chip temperature span also indicates that reducing the overall package temperature did not eliminate the localized thermal nonuniformity generated by the heterogeneous power distribution. The result therefore demonstrated both the effectiveness of embedded liquid cooling and the value of retaining an additional localized heat-transfer mechanism for high-power regions.

Implications for Final Model Development

The intermediate package study reinforced the architectural trends identified by the preliminary single-chip models under a substantially higher and spatially heterogeneous heat load.

Conventional external cooling became increasingly dependent on extreme effective heat-rejection area and remained above the 100 °C design objective even at the largest area ratio investigated. Direct jet impingement continued to provide strong surface cooling: increasing inlet velocity from 1 to 2 m/s reduced maximum temperature from approximately 401 to 365 K, with the 2 and 3 m/s cases remaining below 100 °C. Its principal limitation remained physical access to buried heat-generating regions.

Embedded microchannels provided distributed cooling directly adjacent to the package heat sources. At a total coolant flow rate of 9 g/s, the microchannel model reduced maximum temperature to approximately 381 K (107 °C), substantially below the strongest external-cooling case. The model nevertheless retained a temperature span of approximately 81 K, indicating that distributed cooling alone did not fully suppress the hotspot associated with the heterogeneous power map.

These results provided the final conceptual bridge to the stacked architecture. Microchannels were retained as the primary mechanism for placing coolant within regions that cannot be reached by external cooling, while jet impingement was retained as a complementary localized mechanism where direct access is available. The final model therefore combines distributed internal heat removal with localized impingement enhancement rather than treating either mechanism as a standalone solution.

Table 3.6: Representative results from the intermediate H100-inspired package-level cooling models.

Architecture	Operating condition	$T_{\max}$ (K)	$T_{\max}$ (°C)	ΔT_{chip} (K)
External cooling	$A_{\text{ratio}} = 200$	392	118	88
Microchannels	$\dot{m} = 9 \text{ g/s}$	381	107	81
Jet impingement	$U_{\text{imp}} = 1 \text{ m/s}$	401	127	37
Jet impingement	$U_{\text{imp}} = 2 \text{ m/s}$	365	92	36
Jet impingement	$U_{\text{imp}} = 3 \text{ m/s}$	352	79	34

3.1.4 Transition to the Final Three-Dimensional Hybrid Model

The final modeling stage combined vertically stacked heat-generating domains, embedded microchannels, and an upper jet-impingement chamber into a fully three-dimensional conjugate heat-transfer model. The 3D architecture was developed specifically to resolve the effects that the reduced models could not capture: lateral heat spreading, channel-to-channel coverage, corner hotspots, jet-array interaction, three-dimensional pressure loss, and the combined thermal-hydraulic consequences of varying geometric parameters. The preliminary modeling stages therefore provide a development and verification hierarchy, while the final stacked model provides the thesis's primary quantitative results.

3.2 Final 3D Model Geometry and Heat Load

The three-dimensional computational domain represents three vertically stacked, heat-generating silicon tiers. Each tier has a lateral footprint of $5\text{ mm} \times 5\text{ mm}$ and a thickness of $100\text{ }\mu\text{m}$. The tiers are separated by solid heat-spreading regions and embedded microchannel cooling layers, producing a repeated stack in which heat is removed through internal coolant passages and an upper jet-impingement chamber. A total heat load of 700 W is distributed uniformly among the three active silicon domains, corresponding to approximately 233 W per tier. Normalized by the $5\text{ mm} \times 5\text{ mm}$ projected footprint, the total stack load is approximately 2.8 kW/cm^2 . This is intentionally an extreme steady-state condition used to evaluate the upper thermal capability of the architecture rather than to reproduce a conventional package operating point.

The microchannel layers span the active chip footprint while retaining an uncut solid margin around the channel field. Channel geometry is defined by channel count, height, width, open fraction, pitch, and microchannel edge margin. In the selected configuration, 27 rectangular channels per microchannel layer are distributed across the available width. Each channel is approximately $142\text{ }\mu\text{m}$ wide and $100\text{ }\mu\text{m}$ high, with a pitch of approximately 179

μm and an interchannel wall thickness of approximately $37\ \mu\text{m}$. The microchannel field terminates 0.10 mm from the lateral edge of the active footprint.

Above the stack, a jet plate distributes coolant through an array of circular nozzles into an impingement chamber. Jet geometry is described by nozzle diameter D_j , center-to-center jet pitch p_j , jet pitch ratio p_j/D_j , and nozzle-to-surface standoff distance H . The selected design uses a 4×4 array of $300\ \mu\text{m}$ nozzles, a pitch ratio of 4, a center-to-center pitch of 1.20 mm, and a standoff height of 1.00 mm. The jet and microchannel branches are supplied independently, allowing localized impingement flow to be controlled separately from distributed microchannel flow.

All major dimensions were defined using COMSOL global parameters. Repeated channels and jets were generated using array operations, while Boolean difference and union operations formed the fluid and solid domains. Dynamic geometric selections were used for heat-generating domains, coolant inlets, outlets, walls, and evaluation surfaces so that the intended physical boundaries remained associated with the correct geometry when the channel count or jet array changed. This correction was necessary because early manually indexed boundary selections became inconsistent after parametric geometry rebuilds.

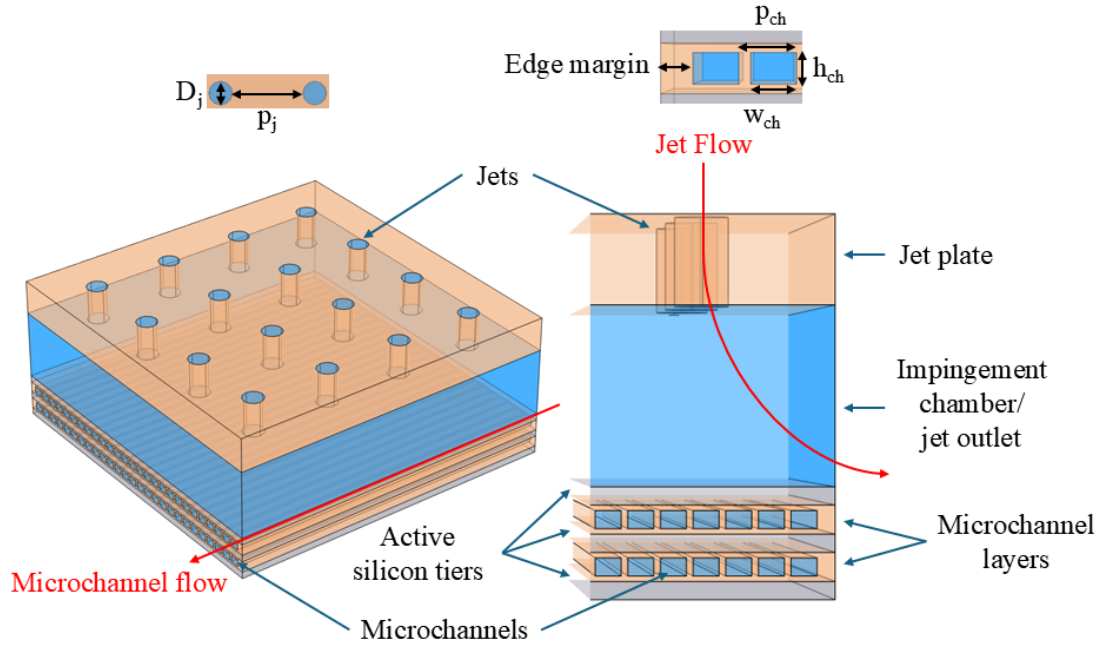


Figure 3.3: Final three-dimensional hybrid cooling model showing isometric view of the complete computational architecture and cutaway view showing the three heat-generating silicon tiers, two embedded microchannel cooling layers, and upper jet-impingement region. Principal geometric parameters and coolant-flow directions are indicated.

3.3 Governing Equations and Performance Metrics

The computational model solves conjugate heat transfer between solid domains and single-phase water flow. Heat conduction within solids is governed by Fourier conduction with volumetric heat generation, while the coolant satisfies conservation of mass, momentum, and energy. Temperature-dependent thermophysical properties are evaluated within COMSOL. Reduced calculations used for screening and analytical comparison employ representative water properties and are identified as estimates.

For steady heat conduction in the solid domains,

$$\nabla \cdot (k \nabla T) + q''' = 0. \quad (3.13)$$

Conservation of mass is

$$\nabla \cdot (\rho \mathbf{u}) = 0, \quad (3.14)$$

and momentum conservation is represented by

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nabla \cdot \boldsymbol{\tau}. \quad (3.15)$$

The steady fluid-energy equation, including the viscous-dissipation contribution enabled through the Nonisothermal Flow coupling, is represented as

$$\rho c_p(\mathbf{u} \cdot \nabla T) = \nabla \cdot (k \nabla T) + Q_{vd}, \quad (3.16)$$

where Q_{vd} represents conversion of mechanical energy to internal energy by viscous stresses. Temperature and heat flux are coupled continuously across the solid-fluid interfaces.

The principal performance metrics are maximum chip temperature, volume-averaged chip temperature, chip temperature span, chip-to-bulk-inlet thermal resistance, branch pressure drop, ideal hydraulic power, mass conservation, and energy conservation. Temperature span is

$$\Delta T_{chip} = T_{max} - T_{min}, \quad (3.17)$$

evaluated over the heat-generating silicon domains. Maximum thermal resistance is

$$R_{th,max} = \frac{T_{max} - T_{bulk,in}}{Q_{gen}}, \quad (3.18)$$

and ideal hydraulic power is

$$P_{hyd} = \Delta P \dot{V} = \Delta P \frac{\dot{m}}{\rho}. \quad (3.19)$$

For interpretation of the selected design, a maximum chip temperature of 100 °C was adopted as an engineering comparison target for the present study. This value is used as a design criterion rather than as a manufacturer-specified junction-temperature limit. At an inlet temperature of 293.15 K and an imposed heat load of 700 W, the corresponding maximum allowable effective chip-to-inlet thermal resistance is

$$R_{\text{th,target}} = \frac{373.15 - 293.15}{700} \approx 0.1143 \text{ K/W}. \quad (3.20)$$

This target provides a direct resistance-scale criterion against which the selected configuration can be evaluated.

For a rectangular channel,

$$D_h = \frac{2w_{\text{ch}}h_{\text{ch}}}{w_{\text{ch}} + h_{\text{ch}}}. \quad (3.21)$$

Reference Reynolds numbers are calculated from mean velocity and hydraulic diameter to assess flow-regime applicability. Preliminary constant-property energy screening uses

$$\dot{Q}_{\text{coolant}} = \dot{m}c_p(T_{\text{bulk,out}} - T_{\text{bulk,in}}), \quad (3.22)$$

while final energy conservation is evaluated using integrated inlet and outlet enthalpy fluxes from COMSOL.

3.4 Materials and Thermophysical Properties

The final three-dimensional model contains silicon heat-generating tiers, copper heat-spreading and cooling-structure regions, and liquid water in the microchannel and jet-flow domains. Material properties were assigned using the COMSOL Multiphysics material library.

Silicon was modeled with a density of 2329 kg/m^3 , a constant-pressure heat capacity of 700 J/(kg K) , and an isotropic thermal conductivity of 130 W/(m K) . Copper was modeled with a density of 8960 kg/m^3 , a constant-pressure heat capacity of 385 J/(kg K) , and an isotropic thermal conductivity of 400 W/(m K) .

Water was used as the coolant in both the microchannel and jet branches. Temperature-dependent water properties from the COMSOL material library were used in the numerical

solution. Density, dynamic viscosity, specific heat, and thermal conductivity were therefore evaluated as functions of local temperature rather than being assigned as constant values. This treatment is important under the imposed heat load because the coolant experiences substantial spatial temperature variation.

Representative constant water properties were used only for reduced analytical calculations and post-processing estimates, including Reynolds number, analytical pressure drop, hydraulic power, and thermal-resistance consistency calculations. These representative values were not substituted for the temperature-dependent properties used in the COMSOL solution.

Table 3.7: Principal thermophysical properties used in the final computational model.

Material	Density	Specific heat	Thermal conductivity / viscosity	Property treatment
Silicon	2329 kg/m ³	700 J/(kg K)	$k = 130 \text{ W/(m K)}$	Constant; COMSOL material library
Copper	8960 kg/m ³	385 J/(kg K)	$k = 400 \text{ W/(m K)}$	Constant; COMSOL material library
Water	$\rho(T)$	$c_p(T)$	$k(T); \mu(T)$	Temperature dependent; COMSOL material library

3.5 Physics Interfaces and Flow Model

Conjugate heat transfer was modeled using the COMSOL Heat Transfer in Solids and Fluids and Laminar Flow physics interfaces. Thermal conduction was solved throughout the solid domains, while forced convection was solved within the microchannel, nozzle, and impingement-chamber fluid domains. No-slip velocity conditions were applied at solid-fluid walls, with thermal continuity enforced at the conjugate interfaces. The water coolant was treated as incompressible and single phase. Buoyancy effects were neglected because forced

convection dominates the momentum and energy transport at the imposed coolant flow rates.

Because the nominal microchannel Reynolds number of the selected configuration is approximately 4,200, the sensitivity of the final result to the assumed flow formulation was examined separately using COMSOL's Turbulent Flow, SST interface. The purpose of this calculation was not to replace the Laminar Flow formulation used throughout the parameter studies, but to determine whether introduction of a RANS turbulence model materially altered the principal thermal and hydraulic conclusions at the highest-flow selected operating condition.

An exploratory SST solution was obtained for the selected normal-mesh geometry using the converged laminar solution as the initial flow and temperature field. The SST formulation also requires a reciprocal wall-distance field for its near-wall treatment. A subsequent attempt to reproduce the calculation using a fully rebuilt wall-distance initialization and stationary SST solution did not achieve convergence within the available computational budget. The SST result is therefore reported only as a model-form sensitivity indicator and is not treated as a validated replacement for the primary laminar solution. The quantitative comparison and its limitations are presented in Section 4.8.6.

3.6 Boundary and Initial Conditions

Water enters the microchannel and jet branches at 293.15 K. The two coolant branches are prescribed independently so that the microchannel and jet flow rates can be varied separately.

For the selected configuration, the microchannel branch receives a total mass flow rate of 25 g/s. The jet branch receives a total mass flow rate of 4 g/s distributed equally among the 16 nozzle inlet boundaries, corresponding to 0.25 g/s per jet. During the variable-count JPR study, the same fixed-total-flow formulation was used so that the prescribed jet-branch mass flow remained 4 g/s as nozzle count changed.

Both coolant branches discharge through static-pressure outlet conditions. The microchannel and jet outlet boundaries are each prescribed at

$$p_0 = 0 \text{ Pa} \quad (3.23)$$

gauge pressure. The Laminar Flow interface uses a reference pressure of 1 atm, such that the corresponding absolute pressure is obtained from

$$p_{\text{abs}} = p + p_{\text{ref}}. \quad (3.24)$$

Backflow suppression is enabled at both outlet conditions.

No-slip velocity conditions are applied at solid-fluid interfaces, while thermal continuity between solid and fluid domains is imposed through the conjugate Heat Transfer in Solids and Fluids formulation. Exterior solid boundaries that do not form part of an inlet, outlet, or solid-fluid heat-transfer interface are modeled as thermally insulated. This treatment includes the lateral and bottom exterior surfaces and prevents heat rejection through unmodeled external cooling paths.

The total imposed heat generation is 700 W, distributed uniformly among the three active silicon tiers. Each tier therefore dissipates approximately 233.3 W. For a tier volume of $5 \text{ mm} \times 5 \text{ mm} \times 100 \text{ } \mu\text{m}$, the corresponding uniform volumetric heat-generation rate is approximately

$$q''' \approx 9.33 \times 10^{10} \text{ W/m}^3. \quad (3.25)$$

The simulations are stationary. Initial values are used only to initialize the nonlinear solution and do not represent a physical transient condition. The fluid domains are initialized with zero velocity and zero gauge pressure,

$$\mathbf{u}_0 = 0, \quad p_0 = 0 \text{ Pa}, \quad (3.26)$$

while all heat-transfer domains are initialized at

$$T_0 = 293.15 \text{ K.} \quad (3.27)$$

Dynamic named selections are used for the channel inlet, channel outlet, jet inlet, jet outlet, heat-generating regions, and other geometry-dependent features so that the intended boundary assignments remain associated with the correct physical surfaces following parametric geometry rebuilds.

Table 3.8: Principal boundary and initial conditions used in the selected configuration.

Condition	Value	Implementation
Total heat generation	700 W	Uniform volumetric heating across three active silicon tiers
Heat generation per tier	233.3 W	Approximately $9.33 \times 10^{10} \text{ W/m}^3$
Microchannel inlet	25 g/s total	Mass-flow inlet to the microchannel branch
Jet inlet	4 g/s total	Equal flow among 16 nozzles; 0.25 g/s per jet
Coolant inlet temperature	293.15 K	Applied to both coolant branches
Microchannel outlet	0 Pa gauge	Static-pressure outlet with backflow suppression
Jet outlet	0 Pa gauge	Static-pressure outlet with backflow suppression
Flow reference pressure	1 atm	Used to obtain absolute pressure from $p_{\text{abs}} = p + p_{\text{ref}}$
Solid-fluid velocity condition	No slip	Applied at all wetted solid surfaces
Solid-fluid thermal condition	Thermal continuity	Conjugate heat transfer
Exterior thermal boundaries	Adiabatic	Thermal insulation on exterior surfaces not participating in coolant heat transfer, including the bottom surface
Initial temperature	293.15 K	Applied to all Heat Transfer domains
Initial velocity	0 m/s	Applied to all fluid domains
Initial gauge pressure	0 Pa	Applied to all fluid domains

3.7 Baseline and Selected Geometric Configurations

Two principal configurations are referenced in the results. The baseline configuration is the first fully converged model after geometric selections and coolant boundary assignments were repaired. It is used as the reference for cumulative improvement. The selected hybrid configuration is a separate final simulation combining the preferred values identified from the individual microchannel and jet-array studies; it is not a single point from one sweep.

This distinction matters because the earlier microchannel sweeps used an auxiliary jet configuration different from the later fixed-total-flow JPR formulation. Their absolute temperatures are therefore used for internally consistent sensitivity comparisons rather than as sequential intermediate states that can be combined algebraically to obtain the selected result.

Table 3.9: Baseline configuration.

Parameter	Baseline value
Chip footprint	5 mm $\times$ 5 mm
Heat-generating silicon tiers	3
Total heat generation	700 W
Microchannel layers	2
Microchannel count per layer	2
Total parallel microchannels	4
Channel width	2.16 mm
Channel height	100 μ m
Channel pitch	2.64 mm
Interchannel wall thickness	480 μ m
Open fraction	0.90
Microchannel edge margin	0.10 mm
Total microchannel mass flow rate	10 g/s
Jet array	2 $\times$ 2
Jet diameter	300 μ m
Jet pitch ratio	8
Jet pitch	2.40 mm
Standoff height	1.00 mm
Total jet mass flow rate	9 g/s
Per-jet mass flow rate	2.25 g/s

Table 3.10: Selected configuration.

Parameter	Selected value
Chip footprint	5 mm $\times$ 5 mm
Heat-generating silicon tiers	3
Total heat generation	700 W
Microchannel layers	2
Microchannel count per layer	27
Total parallel microchannels	54
Channel width	142.2 μm
Channel height	100 μm
Channel pitch	179.1 μm
Interchannel wall thickness	36.9 μm
Open fraction	0.80
Microchannel edge margin	0.10 mm
Total microchannel mass flow rate	25 g/s
Jet array	4 $\times$ 4
Jet diameter	300 μm
Jet pitch ratio	4
Jet pitch	1.20 mm
Standoff height	1.00 mm
Total jet mass flow rate	4 g/s
Per-jet mass flow rate	0.25 g/s

3.8 Parameter Studies

A sequence of parametric studies was conducted to determine the influence of geometric and operating variables on maximum chip temperature, temperature uniformity, hydraulic loss, and geometric practicality. Each study varied one principal parameter while the remaining inputs were held fixed at the values associated with that modeling phase. The studies were performed in two stages because the jet-inlet formulation was revised during model development.

The first stage examined microchannel mass flow rate, channel count, microchannel edge margin, and open fraction using an auxiliary jet configuration. The second stage examined jet pitch ratio (JPR) and array density using a fixed-total-flow formulation in which the total jet mass flow remained constant at 4 g/s as nozzle count changed. Following completion of the individual parameter studies, the preferred parameter values were combined and solved independently as the selected configuration.

Table 3.11: Parameter studies and principal fixed conditions.

Study	Values	Key fixed conditions	Purpose
Microchannel flow	9, 16, 25, 36 g/s	$N_{ch} = 2$ per layer; OF = 0.90; edge margin = 0.10 mm; auxiliary jets: 2×2 , JPR 8, 9 g/s total	Thermal-hydraulic effect of microchannel flow rate
Channel count	8, 27, 64 per layer	$\dot{m}_{mc} = 10$ g/s; OF = 0.90; edge margin = 0.10 mm; auxiliary jets: 2×2 , JPR 8, 9 g/s total	Cooling coverage versus pressure loss and remaining wall thickness
Edge margin	0.50, 0.25, 0.10, 0.05 mm	$N_{ch} = 2$ per layer; $\dot{m}_{mc} = 10$ g/s; OF = 0.90; auxiliary jets: 2×2 , JPR 8, 9 g/s total	Effect of the uncooled perimeter on edge and corner hotspots
Open fraction	0.60, 0.80, 0.95	$N_{ch} = 8$ per layer; $\dot{m}_{mc} = 10$ g/s; edge margin = 0.10 mm; auxiliary jets: 2×2 , JPR 8, 9 g/s total	Flow area versus wall thickness, temperature, and pressure loss
JPR / array density	2, 4, 6, 9	$N_{ch} = 2$ per layer; OF = 0.90; $\dot{m}_{mc} = 10$ g/s; $D_j = 300 \mu\text{m}$; $\dot{m}_{jet} = 4$ g/s total	Jet coverage, per-jet momentum, uniformity, and pressure loss

The fixed conditions in Table 3.11 are important when interpreting the individual sweeps. In particular, the 27-channel case in the channel-count study was evaluated at an open fraction of 0.90, whereas the final selected configuration combines 27 channels per layer with an open fraction of 0.80. Likewise, the open-fraction study was performed with eight channels per layer rather than 27. The wall thicknesses and absolute temperatures reported within the individual sweeps therefore belong to those specific study configurations and should not be transferred directly to the final combined geometry. The preferred values identified from the separate studies were subsequently combined and solved independently as the selected configuration.

For the first four studies, the auxiliary jet configuration used a JPR of 8, a 2×2 nozzle array, and a total jet-branch mass flow rate of 9 g/s. These studies are therefore interpreted as internally consistent microchannel sensitivity studies rather than as direct absolute comparisons with the later jet-array study or selected configuration.

For the JPR study, the usable impingement footprint was populated with as many 300 μm -diameter nozzles as permitted by the prescribed pitch. Total jet mass flow remained fixed at 4 g/s. Consequently, changes in JPR also changed nozzle count, individual nozzle mass flow, jet velocity, and jet Reynolds number. The resulting study represents a system-level investigation of JPR and array density rather than an isolated fixed-nozzle-count pitch study.

The selected configuration combined a microchannel mass flow rate of 25 g/s, 27 channels per microchannel layer, a microchannel edge margin of 0.10 mm, an open fraction of 0.80, and a JPR of 4. The selected jet geometry consisted of a 4×4 array of 300 μm -diameter nozzles with a total jet-branch mass flow rate of 4 g/s. This configuration was solved independently after completion of the individual parameter studies.

3.9 Mesh, Solver, and Numerical Implementation

All three-dimensional simulations were performed using COMSOL Multiphysics 6.3.0.420. The final studies employed a stationary nonlinear formulation coupling incompressible fluid momentum and continuity with conjugate heat transfer in the solid and fluid domains. Geometric and operating inputs were defined using global parameters, and dynamic geometric selections were used where practical to preserve material, physics, boundary-condition, and post-processing assignments during parametric geometry rebuilds.

The computational domain was discretized using a mixed-element mesh containing tetrahedral, prismatic, and pyramidal elements. The normal mesh used for the primary parameter studies and selected configuration contained 852,151 domain elements and 191,028 mesh vertices. The domain mesh consisted of 719,011 tetrahedral elements, 132,244 prismatic elements, and 896 pyramidal elements. Mesh quality was evaluated using the COMSOL skewness quality measure. The normal mesh had an average element quality of 0.654 and a minimum element quality of 0.100. The mixed-element formulation accommodated the narrow coolant passages, thin solid features, jet geometry, and larger three-dimensional regions of the computational domain.

Table 3.12: Mesh levels used for the selected-configuration mesh-sensitivity study.

Mesh	Domain elements	Average quality	Minimum quality
Coarse	747,442	0.5795	0.0937
Normal	852,151	0.6541	0.0997
Fine	2,745,247	0.6550	0.0990

A three-level mesh-sensitivity study was performed on the selected configuration using coarse, normal, and fine meshes containing approximately 7.47×10^5 , 8.52×10^5 , and 2.75×10^6 domain elements, respectively. Maximum chip temperature, chip temperature span, thermal resistance, and microchannel pressure drop were selected as the principal comparison metrics. The resulting mesh sensitivity is evaluated quantitatively in Chapter 4. Because the thermal and hydraulic quantities did not exhibit identical convergence behavior,

the study is treated as a mesh-sensitivity assessment rather than as evidence of complete mesh independence.

The stationary nonlinear system was solved using a segregated solution strategy. Velocity and pressure were solved as the first segregated group, followed by temperature as a separate group. The overall stationary solver used a relative tolerance of 0.001 with automatic linearity detection. In the selected normal-mesh solution, approximately 1.24×10^6 degrees of freedom were solved, in addition to approximately 2.44×10^5 internal degrees of freedom.

The fluid-flow linear system used the GMRES iterative solver with algebraic multigrid (AMG) preconditioning. GMRES was configured with a restart interval of 50 iterations, a residual tolerance of 0.01, and a maximum of 1000 iterations. Nonlinear-based error estimation, nonlinear variable scaling, and GCRO-DR subspace recycling were enabled. The multigrid implementation employed iterative pre- and post-smoothing operations and a direct coarse-grid solution.

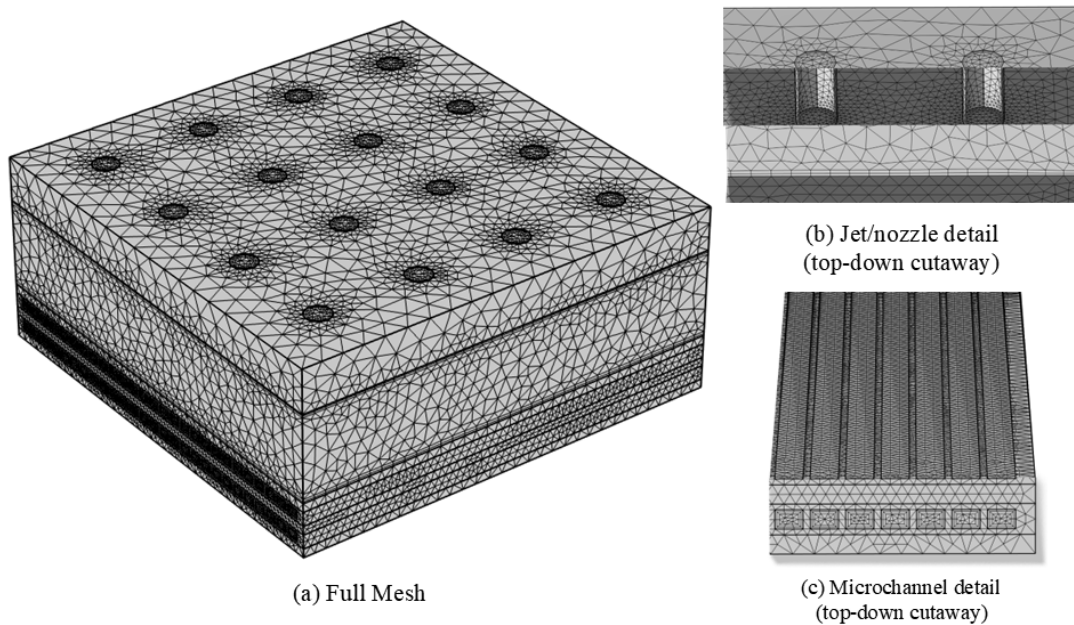


Figure 3.4: Normal mesh used for the selected three-dimensional configuration: (a) complete computational domain and representative refinement in (b) the jet/nozzle region and (c) the embedded microchannel region. The normal mesh contains 852,151 domain elements.

3.10 Model Assumptions and Limitations

The computational model was developed primarily for comparative evaluation of geometric and operating parameters within the proposed hybrid jet-microchannel cooling architecture. Several simplifying assumptions were therefore introduced to maintain computational feasibility while preserving the dominant thermal-fluid behavior.

Water was modeled as an incompressible, single-phase liquid. Temperature-dependent water properties were obtained from the COMSOL material library, while the reduced analytical calculations used representative properties appropriate to the corresponding operating range. Phase change, cavitation, dissolved gases, and two-phase transport were not explicitly modeled.

The validity of the single-phase assumption was evaluated for the selected configuration by comparing the maximum coolant temperature with the saturation temperature corresponding to the local absolute pressure. The Laminar Flow interface used a reference pressure of 1 atm, such that the solved pressure field represented gauge pressure and absolute pressure was obtained from

$$p_{\text{abs}} = p + p_{\text{ref}}. \quad (3.28)$$

The resulting saturation-margin assessment is presented in Chapter 4.

The applied heat load was steady and spatially uniform within each active silicon tier. Transient workload variation, moving hotspots, and transistor-level power maps were not included. The package geometry was also simplified relative to a fabricated three-dimensional integrated device. Microbumps, through-silicon vias, underfill heterogeneity, individual bonding layers, surface roughness, and thermo-mechanical deformation were not explicitly resolved unless represented by an equivalent bulk solid region. The resulting model therefore represents the thermal-fluid behavior of the proposed cooling architecture rather than a complete transistor-to-package digital twin.

Thermal contact resistance was not imposed as a separate interfacial boundary condition within the final three-dimensional model. The simplified solid-solid interfaces were treated using thermal continuity. Consequently, the reported selected-configuration temperatures represent an idealized bonded-interface condition and do not include the additional junction-to-coolant resistance that would be introduced by finite bond, underfill, microbump, hybrid-bond, or thermal-interface resistances in a fabricated package. Interface-resistance sensitivity and degradation are therefore treated as package-level extensions beyond the scope of the present architecture-focused study.

For the final jet-array study, equal mass flow was prescribed at each nozzle inlet while total jet-branch mass flow remained fixed. This formulation provides controlled comparison of array coverage and per-jet momentum but does not resolve flow maldistribution generated by an upstream supply plenum. The earlier microchannel parameter studies used a different auxiliary jet configuration and are consequently interpreted as internally consistent sensitivity studies rather than as direct absolute comparisons with the final JPR study.

All primary parameter studies were solved using the Laminar Flow interface to maintain a consistent physics formulation across the investigated design space. Reference Reynolds-number calculations show that the selected high-flow microchannel configuration extends beyond the conventional fully laminar regime. An exploratory SST turbulence-model calculation was therefore used to assess the potential model-form sensitivity of the selected configuration. The comparison is reported in Chapter 4 and is used as a sensitivity indicator rather than as a replacement for the laminar parameter-study results. Because a subsequently rebuilt SST solution with explicit reciprocal wall-distance initialization did not converge, the turbulent comparison is not treated as a fully independent validation case.

Additional numerical uncertainty arises from discretization of the narrow microchannels, thin interchannel walls, developing flow regions, jet stagnation zones, and abrupt inlet and outlet transitions. The mesh-sensitivity study consequently considers both thermal and hydraulic quantities rather than relying exclusively on maximum temperature. These

assumptions and limitations permit interpretation of the parameter trends and selected design while constraining the extent to which the reported absolute values can be treated as fully validated predictions of a fabricated package.

3.11 Analytical and Numerical Validation Methods

The numerical model was evaluated using complementary conservation checks, reduced-order analytical calculations, mesh sensitivity, thermodynamic consistency checks, and comparison with published experimental results. These methods were selected to test both numerical consistency and the physical magnitude of the predicted thermal and hydraulic behavior without requiring the reduced-order calculations to reproduce the complete three-dimensional conjugate model.

3.11.1 Mass and Energy Conservation

Global mass conservation was evaluated by integrating the mass flux across the coolant inlet and outlet boundaries. The relative mass imbalance was calculated from the difference between the total inlet and outlet mass flow normalized by the prescribed inlet flow.

A preliminary sensible-energy estimate was obtained from

$$\dot{Q}_{\text{sens}} = \dot{m} c_p (T_{\text{bulk,out}} - T_{\text{bulk,in}}), \quad (3.29)$$

where representative constant-pressure specific heat was used for the reduced calculation. The corresponding expected bulk coolant temperature rise was estimated as

$$\Delta T_{\text{bulk}} = \frac{\dot{Q}}{\dot{m} c_p}. \quad (3.30)$$

The final numerical energy balance was evaluated directly using the COMSOL Heat Transfer interface total-energy-flux variable, `ht.nteflux`. Surface integrations were performed independently over the microchannel inlet, microchannel outlet, jet inlet, and jet

outlet boundaries. Because the reported flux is referenced to the outward surface normal, inlet contributions are negative and outlet contributions are positive. The net energy gained by the coolant was calculated from

$$\dot{Q}_{\text{coolant}} = \dot{E}_{\text{mc,out}} + \dot{E}_{\text{jet,out}} + \dot{E}_{\text{mc,in}} + \dot{E}_{\text{jet,in}}, \quad (3.31)$$

where the signed inlet and outlet values were retained directly from COMSOL. The resulting energy recovery was compared with the imposed 700 W volumetric heat load. This direct flux balance avoids the constant-property approximation used in the preliminary sensible-energy screening calculation.

3.11.2 Microchannel Pressure-Drop Estimate

An independent pressure-drop estimate was developed for the selected microchannel geometry to assess the physical scale of the numerically predicted hydraulic resistance. The selected configuration contains two microchannel layers with 27 channels per layer, corresponding to 54 nominally parallel flow passages. The prescribed total microchannel mass flow rate of 25 g/s was assumed to distribute uniformly among these passages for the reduced-order calculation.

Each channel has a width of approximately $142.2 \mu\text{m}$, a height of $100 \mu\text{m}$, and a flow length of 5 mm. The rectangular hydraulic diameter is

$$D_h = \frac{2w_{\text{ch}}h_{\text{ch}}}{w_{\text{ch}} + h_{\text{ch}}}, \quad (3.32)$$

which gives approximately $117.4 \mu\text{m}$ for the selected geometry. The nominal mean velocity in each channel is calculated from

$$U = \frac{\dot{m}_{\text{mc}}}{N_{\text{ch,total}}\rho w_{\text{ch}}h_{\text{ch}}}, \quad (3.33)$$

and the corresponding Reynolds number is

$$Re = \frac{\rho U D_h}{\mu}. \quad (3.34)$$

Because water properties vary appreciably over the temperature range of the selected configuration, representative properties for the reduced-order calculations were evaluated near the mean microchannel bulk temperature rather than at an arbitrary elevated reference temperature. An independent sensible-energy estimate gives a bulk temperature rise of approximately 6.7 K at 25 g/s if the full 700 W heat load is conservatively assigned to the microchannel branch, corresponding to a representative mean bulk temperature of approximately 296.5 K. At this condition, representative water properties of approximately $\rho = 997 \text{ kg/m}^3$ and $\mu = 9.2 \times 10^{-4} \text{ Pa s}$ were used.

These values give a nominal mean channel velocity of approximately 32.7 m/s and a Reynolds number of approximately

$$Re_{ch} \approx 4.2 \times 10^3. \quad (3.35)$$

The numerical COMSOL model itself does not use these constant properties; temperature-dependent water properties from the material model are retained throughout the conjugate heat-transfer solution. The representative values are used only for the reduced analytical calculations.

For comparison with the numerical hydraulic result, the fully developed rectangular-duct relation

$$\Delta P_{\text{lam}} \approx \frac{12\mu L Q_{\text{ch}}}{w_{\text{ch}} h_{\text{ch}}^3 (1 - 0.63 h_{\text{ch}}/w_{\text{ch}})} \quad (3.36)$$

was applied, where Q_{ch} is the volumetric flow rate through one channel. Because the representative Reynolds number lies above the range in which a fully developed laminar approximation can be applied without qualification, this calculation is treated as a reduced physical-consistency check rather than as an exact hydraulic prediction. The relation also

neglects developing-flow behavior and three-dimensional entrance, exit, and geometric effects represented by the COMSOL model. The quantitative comparison is presented in Chapter 4.

3.11.3 Representative One-Dimensional Thermal-Resistance Model

A representative one-dimensional thermal-resistance model was developed as an independent check on the characteristic temperature scale predicted by the three-dimensional COMSOL solution. The reduced model is based on the thermal-ladder framework introduced in Section 3.1.1 but collapses the distributed network into a simplified conduction-convection path. It is not intended to reproduce the local maximum temperature or the complete three-dimensional heat-transfer network.

The solid contribution is represented by through-thickness conduction across one heat-generating silicon tier,

$$R_{\text{Si}} = \frac{L_{\text{Si}}}{k_{\text{Si}} A_{\text{chip}}}, \quad (3.37)$$

where L_{Si} is the silicon-tier thickness, k_{Si} is the silicon thermal conductivity, and A_{chip} is the active chip footprint.

The coolant-side resistance is estimated independently from the selected microchannel geometry and flow condition rather than inferred from the COMSOL temperature solution. For the rectangular microchannels, the hydraulic diameter is

$$D_h = \frac{2w_{\text{ch}}h_{\text{ch}}}{w_{\text{ch}} + h_{\text{ch}}}, \quad (3.38)$$

and the representative Prandtl number is calculated from

$$Pr = \frac{\mu c_p}{k_f}. \quad (3.39)$$

Because the selected operating condition lies above the conventional fully laminar

regime, the Gnielinski correlation is used as a reduced-order engineering estimate of the forced-convection heat-transfer scale [16]. The smooth-channel friction factor is approximated by

$$f = (0.79 \ln Re - 1.64)^{-2}, \quad (3.40)$$

and the corresponding Nusselt number is

$$Nu = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}. \quad (3.41)$$

The representative microchannel heat-transfer coefficient is then

$$h_{mc} = \frac{Nu k_f}{D_h}. \quad (3.42)$$

The total wetted area of the two microchannel layers is estimated from

$$A_{wet,mc} = N_{ch,total} [2(w_{ch} + h_{ch})L_{ch}], \quad (3.43)$$

where $N_{ch,total}$ is the total number of nominally parallel channels across both microchannel layers. The corresponding representative convective resistance is

$$R_{conv,mc} = \frac{1}{h_{mc}A_{wet,mc}}, \quad (3.44)$$

giving the reduced one-dimensional resistance

$$R_{1D} = R_{Si} + R_{conv,mc}. \quad (3.45)$$

To preserve independence from the numerical branch-level energy split, the full imposed heat load is conservatively assigned to the representative microchannel path. The corresponding bulk coolant temperature rise is estimated as

$$\Delta T_{\text{bulk}} = \frac{Q_{\text{gen}}}{\dot{m}_{\text{mc}} c_p}, \quad (3.46)$$

and the representative mean coolant temperature is

$$T_{\text{bulk,mean}} = T_{\text{in}} + \frac{\Delta T_{\text{bulk}}}{2}. \quad (3.47)$$

The characteristic one-dimensional chip temperature is then estimated from

$$T_{1\text{D}} = T_{\text{bulk,mean}} + Q_{\text{gen}} R_{1\text{D}}. \quad (3.48)$$

The resulting temperature is compared with the volume-averaged chip temperature from the three-dimensional numerical model rather than with the localized maximum temperature. This distinction is necessary because the reduced model does not resolve lateral heat spreading, local hotspots, developing-flow effects, spatial coolant-temperature variation, the detailed three-tier thermal ladder, or the parallel heat-removal contribution of the jet-impingement branch. The calculation is therefore used as an independent thermal-scale consistency check rather than as an analytical reproduction of the complete conjugate heat-transfer solution.

3.11.4 Mesh Sensitivity

Mesh sensitivity was evaluated using the coarse, normal, and fine meshes summarized in Table 3.12. Maximum chip temperature, chip temperature span, maximum thermal resistance, and microchannel pressure drop were compared between mesh levels. Particular emphasis was placed on the normal-to-fine comparison because the fine mesh contained more than three times as many domain elements as the normal mesh. The numerical results of this comparison are reported in Chapter 4. Because different output quantities exhibited different levels of mesh sensitivity, no assumption of complete mesh independence was imposed.

3.11.5 Single-Phase Consistency Check

The single-phase water assumption was assessed using a local thermodynamic consistency check. The maximum water temperature was first identified within the coolant domains. Because this maximum occurred at a solid-fluid interface, the local pressure was evaluated immediately within the adjacent fluid domain. Absolute pressure was obtained using Eq. 3.28, and the corresponding water saturation temperature was determined from the local absolute pressure. A positive difference

$$\Delta T_{\text{sub}} = T_{\text{sat}}(p_{\text{abs}}) - T_{\text{water}} \quad (3.49)$$

indicates that the coolant remains locally subcooled at the maximum-temperature location. The numerical saturation margin for the selected configuration is reported in Chapter 4.

3.11.6 Comparison with Published Hybrid-Cooling Performance

As an additional plausibility comparison, the effective thermal-resistance scale obtained from the selected configuration is compared with published hybrid microchannel-jet cooling measurements. Xiao et al. reported a thermal resistance of approximately 0.11 K/W for a different hybrid microchannel-jet architecture [5]. Differences in geometry, heat load, flow conditions, and cooling architecture prevent direct validation against that value; the comparison is therefore used only to assess whether the predicted thermal-resistance scale is consistent with experimentally demonstrated hybrid-cooling performance.

Chapter 4: Results and Discussion

This chapter presents the numerical results of the parametric studies and interprets their thermal-fluid significance. Results and discussion are integrated so that the physical mechanism, hydraulic or geometric penalty, and rationale for each selected design value are considered within the same subsection. Because the early microchannel studies and later jet-array study used different auxiliary jet configurations, comparisons are made primarily within each individual sweep. The selected configuration was subsequently solved independently using the preferred parameter values combined in a single model.

4.1 Baseline Performance

The baseline configuration represents the first fully converged reference model obtained after correction of the geometric selections and coolant boundary assignments. It produced a maximum chip temperature of approximately 477 K (204 °C), a volume-averaged chip temperature of approximately 418 K, and a chip temperature span of approximately 156 K. The corresponding maximum chip-to-bulk-inlet thermal resistance was approximately 0.262 K/W.

The baseline maximum temperature of approximately 477 K occurred at $(x, y, z) = (0, 1.8663, 0)$ mm within the lowest active silicon tier. Its location toward the outer portion of the active footprint is consistent with the limited coolant coverage and comparatively long conduction path present in the baseline geometry. The subsequent edge-margin and channel-density studies therefore evaluate whether extending active cooling toward these peripheral regions can suppress the baseline hotspot.

Global mass flow in the baseline model was conserved to within approximately 0.0002%. A preliminary constant-property sensible-energy calculation recovered approximately 95.4% of the imposed heat load. These preliminary checks were used during model development to identify boundary-condition and solution inconsistencies. More rigorous

conservation and physical-consistency checks were subsequently performed on the selected configuration and are presented in Section 4.8.

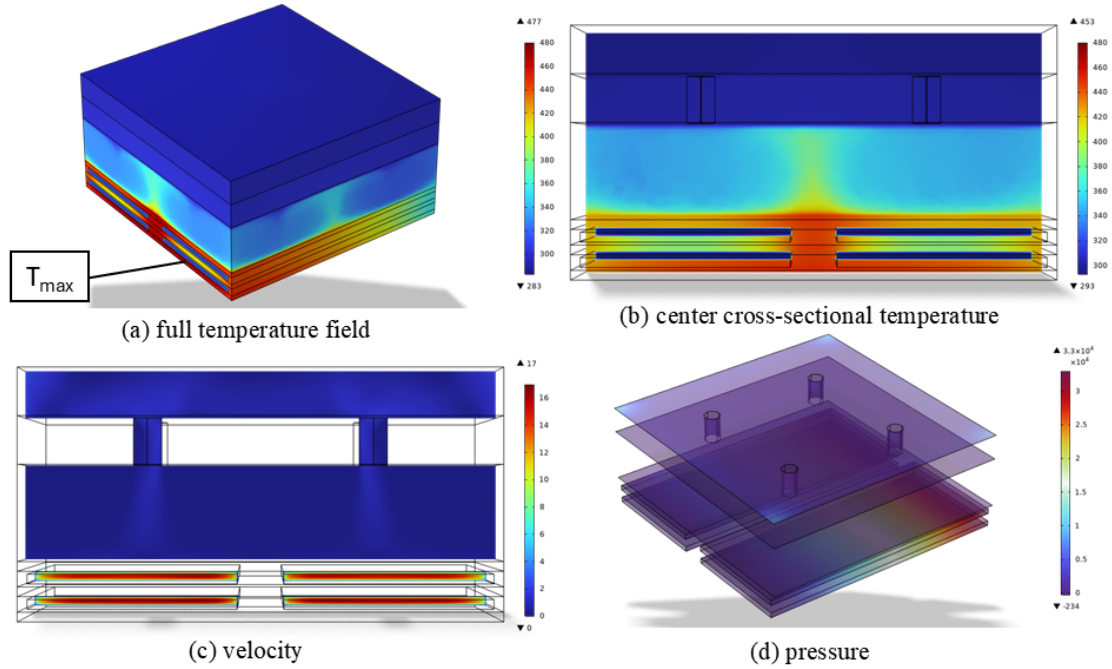


Figure 4.1: Representative fields for the baseline configuration: (a) chip temperature with the maximum-temperature location indicated, (b) cross-sectional temperature through the stacked structure, (c) coolant velocity magnitude, and (d) coolant gauge pressure. Temperature panels use a common 293-480 K scale that is also applied to the selected-configuration comparison in Fig. 4.15.

Table 4.1: Numerical screening metrics for the baseline and selected configurations.

Metric	Baseline	Selected configuration
Absolute mass imbalance	0.0002%	0.0008%
Preliminary constant- c_p energy recovery	95.4%	99.2%
Maximum thermal resistance	0.262 K/W	0.115 K/W
Estimated jet Reynolds number	9,549	1,061

The baseline remains substantially above the 100 °C design objective and exhibits considerable spatial nonuniformity. It therefore provides a useful reference for evaluating whether changes in channel flow, cooling coverage, edge margin, and jet-array configuration address the mechanisms responsible for the hotspot rather than merely reducing the volume-averaged temperature.

4.2 Effect of Microchannel Mass Flow Rate

Increasing microchannel mass flow from 9 to 36 g/s reduced maximum chip temperature from approximately 484 to 421 K, a reduction of approximately 63 K. Average temperature decreased by approximately 59 K and chip temperature span decreased from approximately 159 to 125 K. Channel pressure drop increased from 28.3 to 115.5 kPa, while ideal channel hydraulic power increased from approximately 0.26 to 4.17 W.

Table 4.2: Thermal and hydraulic results of the microchannel mass-flow sweep.

$\dot{m}_{mc}$ (g/s)	T_{max} (K)	T_{avg} (K)	ΔT_{chip} (K)	ΔP_{ch} (kPa)	$R_{th,max}$ (K/W)	P_{hyd} (W)
9	484	423	159	28.3	0.272	0.26
16	457	399	150	54.5	0.234	0.88
25	438	380	139	81.6	0.207	2.05
36	421	364	125	115.5	0.183	4.17

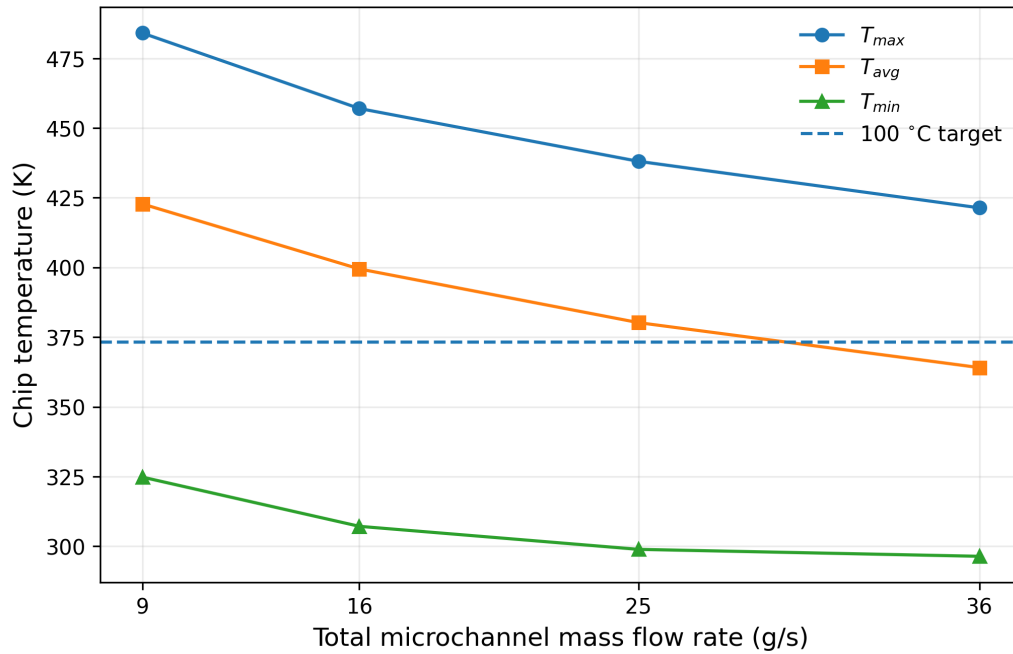


Figure 4.2: Chip temperature metrics versus microchannel mass flow rate.

The thermal improvement is nonlinear. Increasing flow from 9 to 16 g/s reduced maximum temperature by approximately 27 K, while the larger increase from 25 to 36 g/s pro-

duced only an additional 17 K reduction. At higher flow, coolant-side convection represents a smaller fraction of the total junction-to-coolant resistance, causing solid conduction and three-dimensional heat spreading to become increasingly important limitations. Hydraulic demand, however, continues to increase strongly with flow. The relevant design trend is therefore the opposing curvature of thermal improvement and hydraulic demand rather than simply the reduction in temperature with increasing flow [3].

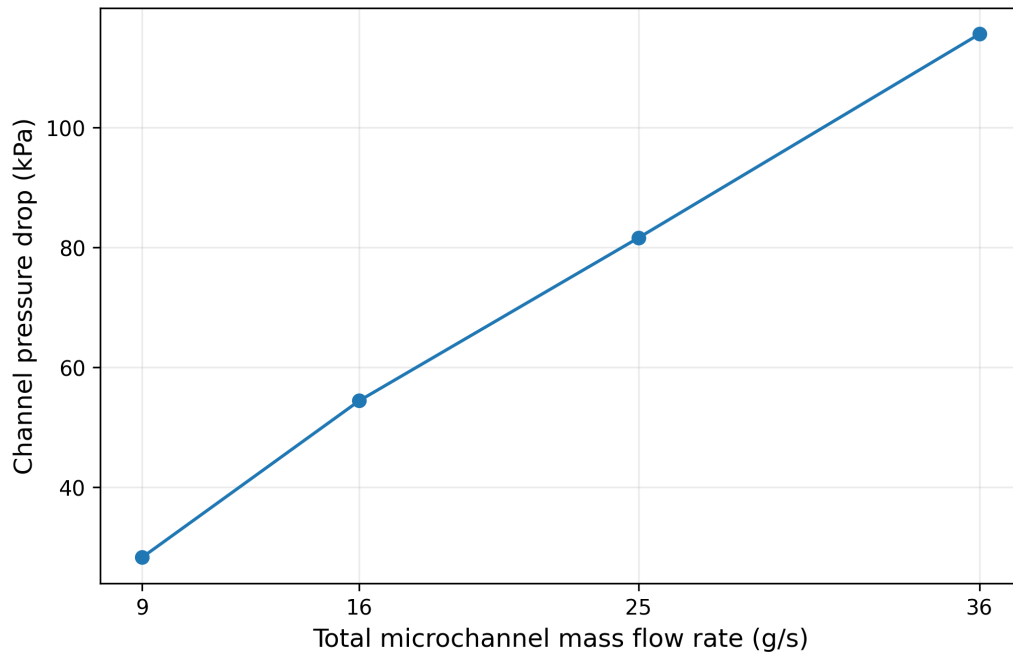


Figure 4.3: Channel pressure drop versus microchannel mass flow rate.

The 25 g/s case was selected as a performance-oriented compromise. Relative to 16 g/s, it reduced maximum temperature by approximately 19 K. Increasing the flow further to 36 g/s produced only another approximately 17 K reduction while approximately doubling the ideal hydraulic power from 2.05 to 4.17 W.

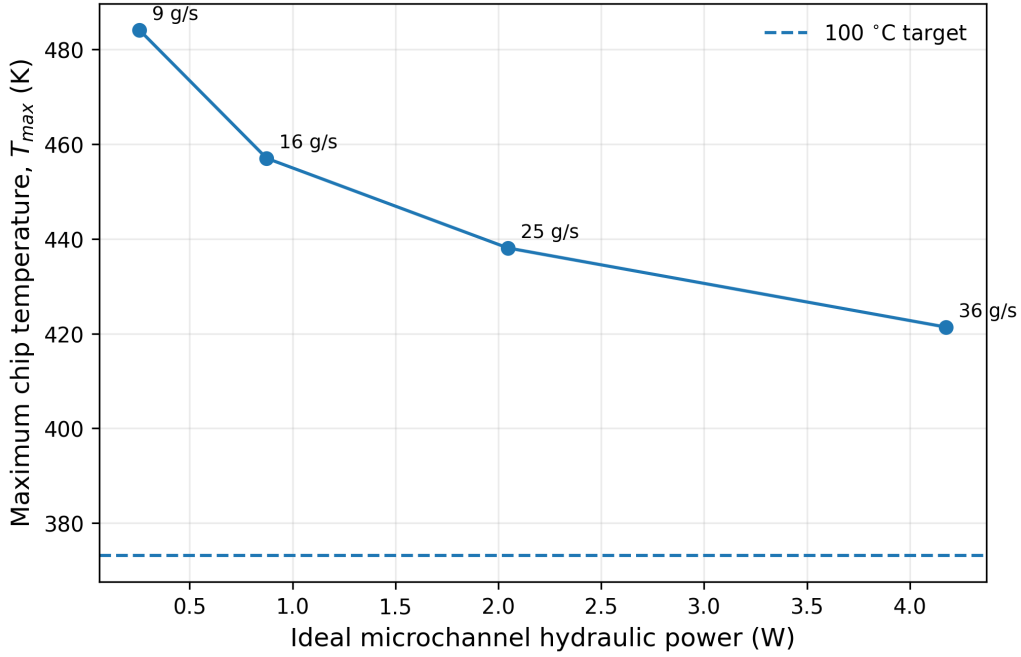


Figure 4.4: Thermal-hydraulic tradeoff for the microchannel mass-flow sweep.

4.3 Effect of Channel Count

Increasing channel count from 8 to 64 reduced maximum chip temperature from approximately 444 to 375 K and reduced chip temperature span from approximately 119 to 66 K. Channel pressure drop increased from 39.7 to approximately 202 kPa, while the interchannel wall thickness decreased from approximately 68.6 to 7.6 μm .

Table 4.3: Thermal, hydraulic, and geometric results of the channel-count sweep.

Channels	T_{max} (K)	ΔT_{chip} (K)	ΔP_{ch} (kPa)	Wall thickness (μm)	Mass imbalance (%)
8	444	119	39.7	68.6	0.006
27	409	88	80.0	18.5	0.390
64	375	66	201.9	7.6	0.007

Increasing channel density distributes cooled surfaces more densely across the footprint and reduces the characteristic solid-conduction distance to a coolant passage. The same change increases hydraulic resistance and removes solid material between adjacent passages. The 64-channel case therefore provides strong thermal performance but represents

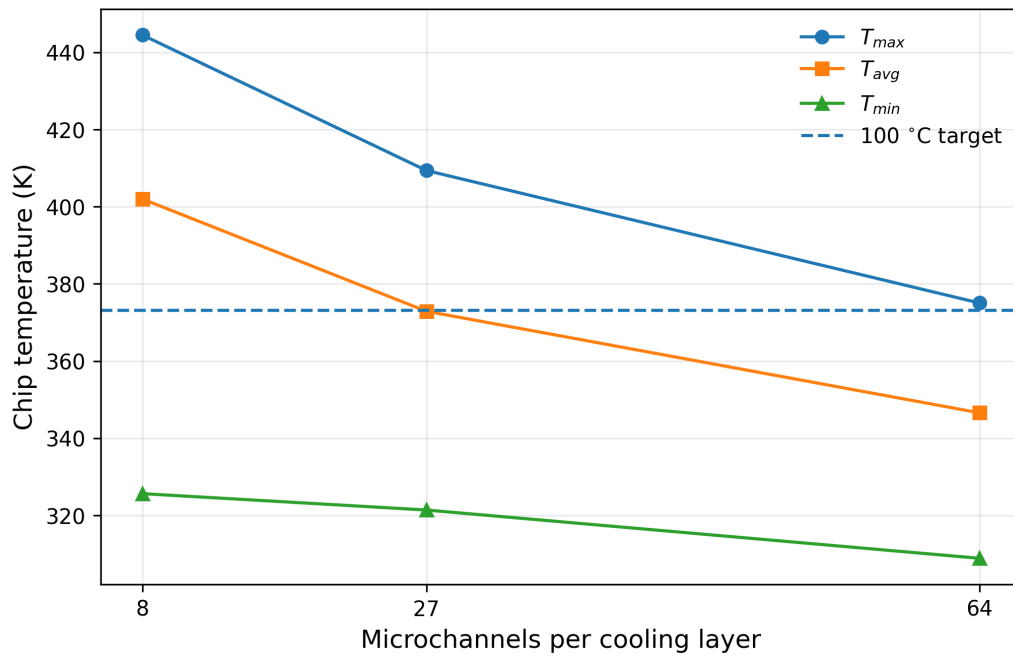


Figure 4.5: Chip temperature metrics versus microchannel count.

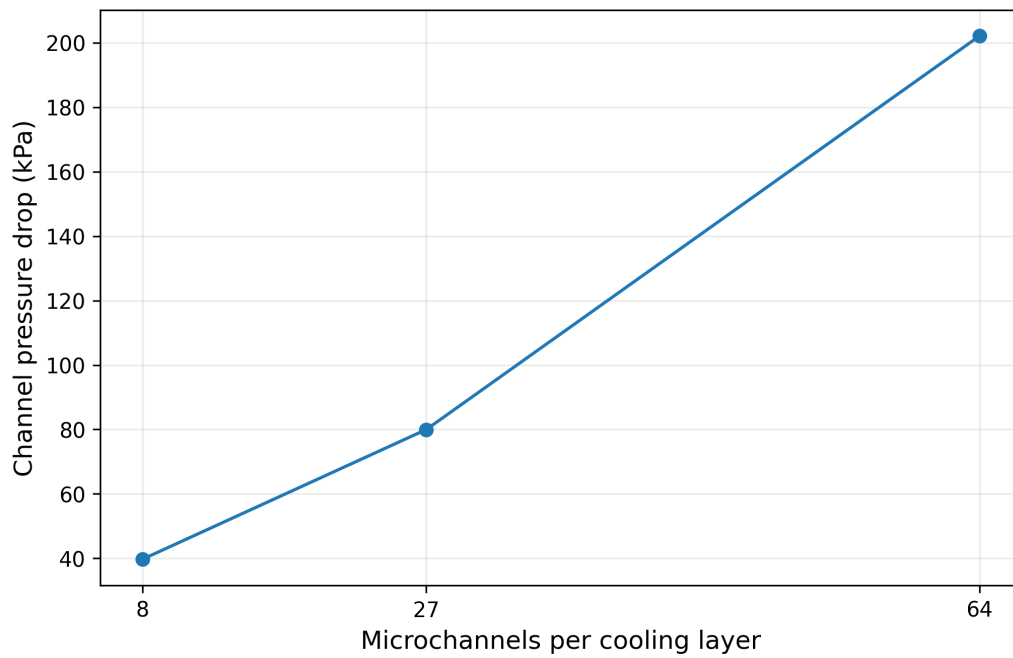


Figure 4.6: Channel pressure drop versus microchannel count.

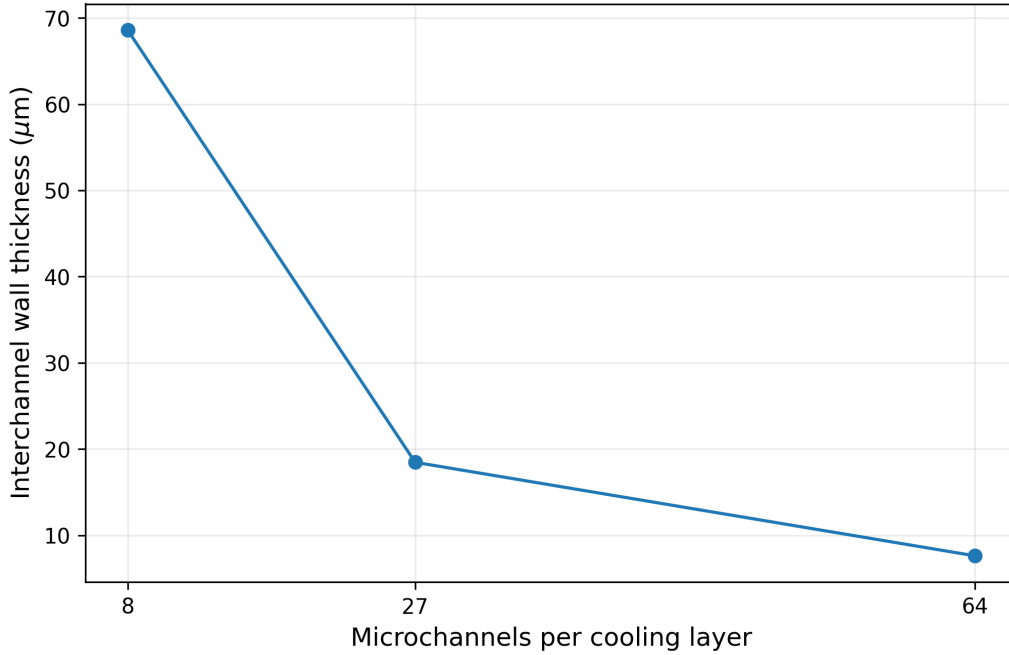


Figure 4.7: Remaining interchannel wall thickness versus microchannel count.

an aggressive physical design: the approximately $7.6 \mu\text{m}$ ligament provides little geometric margin, while the approximately 202 kPa pressure drop is substantial. The 27-channel case was selected as an intermediate design and was subsequently paired with an open fraction of 0.80, producing a final interchannel wall thickness of approximately $36.9 \mu\text{m}$.

The 0.39% mass imbalance in the exported 27-channel sweep case is larger than in neighboring cases and is treated as a numerical-quality observation rather than a physical trend. The independently solved selected configuration exhibited substantially better mass conservation.

4.4 Effect of Microchannel Edge Margin

The microchannel edge margin is the lateral distance between the outer channel field and the edge of the active $5 \times 5 \text{ mm}$ footprint. Heat generated within this uncooled perimeter must conduct laterally before reaching a coolant passage. Reducing the margin from 0.50 to 0.25 mm reduced maximum chip temperature by approximately 44 K. Further reduction to

0.10 mm produced only a small additional improvement, while reduction to 0.05 mm did not further decrease the hotspot temperature.

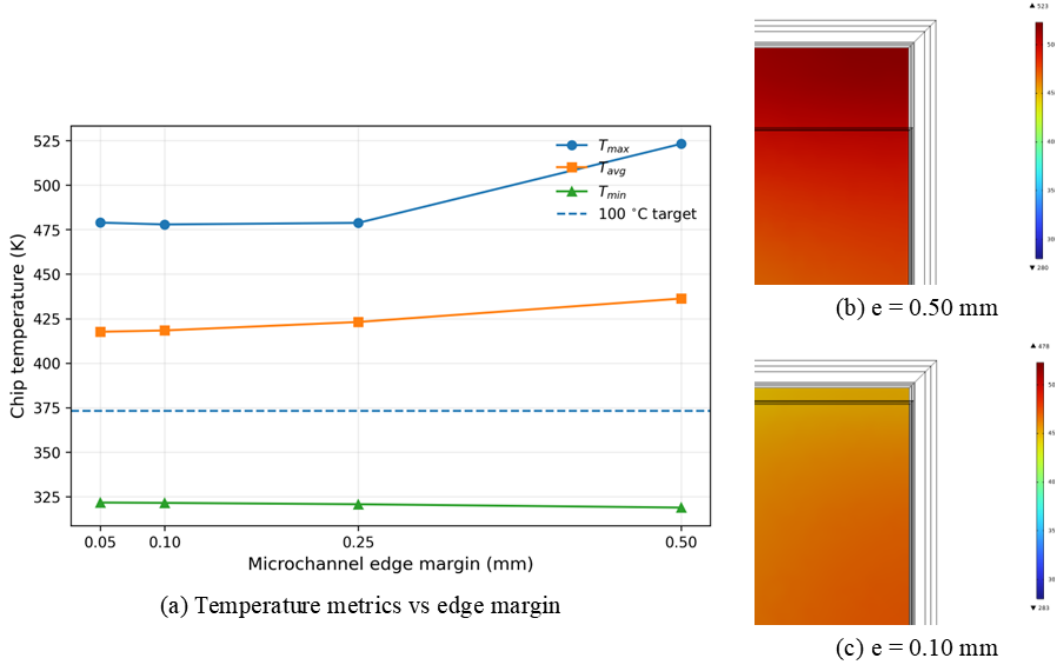


Figure 4.8: Effect of microchannel edge margin on chip temperature and perimeter-hotspot formation: (a) temperature metrics across the edge-margin sweep, (b) representative active-tier temperature field at a 0.50 mm edge margin, and (c) the corresponding field at a 0.10 mm edge margin. Panels (b) and (c) use identical spatial and temperature scales.

The strong initial sensitivity indicates that the large-margin hotspot was controlled primarily by the uncooled perimeter. Once the channel field approached within approximately 0.10 mm of the active-region edge, the lateral conduction path ceased to dominate the thermal resistance and the response plateaued. A 0.10 mm margin was therefore selected because it captured nearly all of the observed thermal benefit while retaining a finite solid perimeter for geometric robustness, meshing, sealing, and fabrication.

4.5 Effect of Channel Open Fraction

Increasing open fraction from 0.60 to 0.80 reduced maximum temperature by approximately 15 K and reduced channel pressure drop by approximately 31%. Increasing the open fraction further to 0.95 reduced maximum temperature by only approximately 1 K. The 0.80 case

produced the smallest chip temperature span, whereas the 0.95 case produced the lowest average temperature and pressure drop.

Table 4.4: Thermal, hydraulic, and geometric results of the channel open-fraction sweep.

Open fraction	T_{max} (K)	T_{avg} (K)	ΔT_{chip} (K)	ΔP_{ch} (kPa)	Wall thickness (μm)
0.60	462	420	138	64.4	274.3
0.80	447	408	120	44.3	137.1
0.95	446	401	121	37.2	34.3

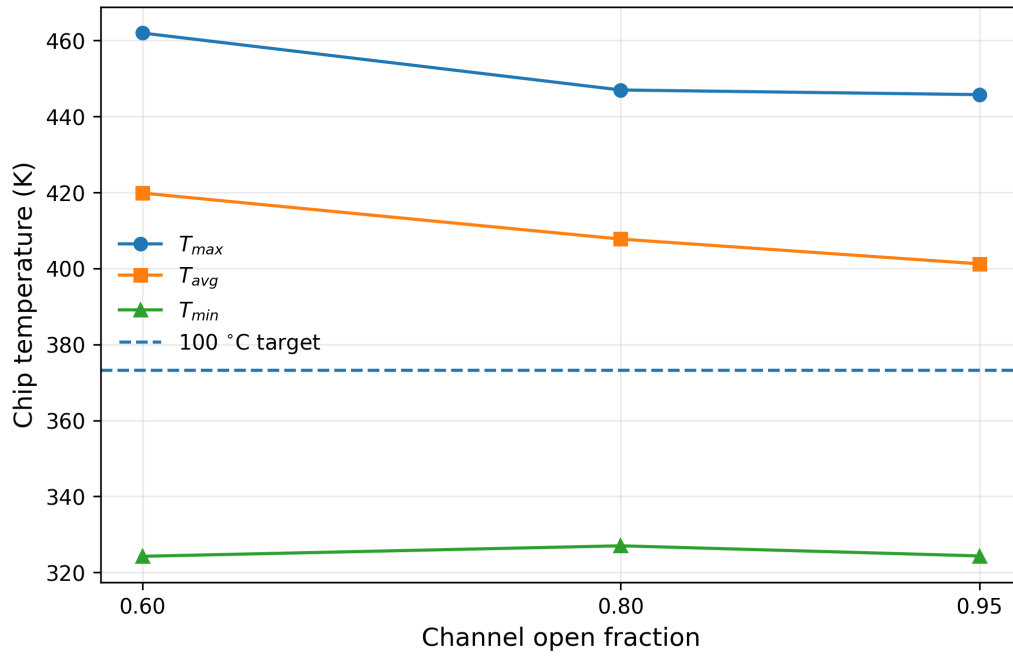


Figure 4.9: Chip temperature metrics versus channel open fraction.

Increasing open fraction increases fluid passage area and reduces hydraulic resistance, but it also removes solid material between adjacent passages. The largest combined thermal and hydraulic improvement occurs between 0.60 and 0.80. Beyond 0.80, improvement in maximum temperature becomes small while the remaining solid ligament continues to decrease. An open fraction of 0.80 was therefore selected as a compromise among peak temperature, temperature uniformity, pressure drop, and geometric robustness.

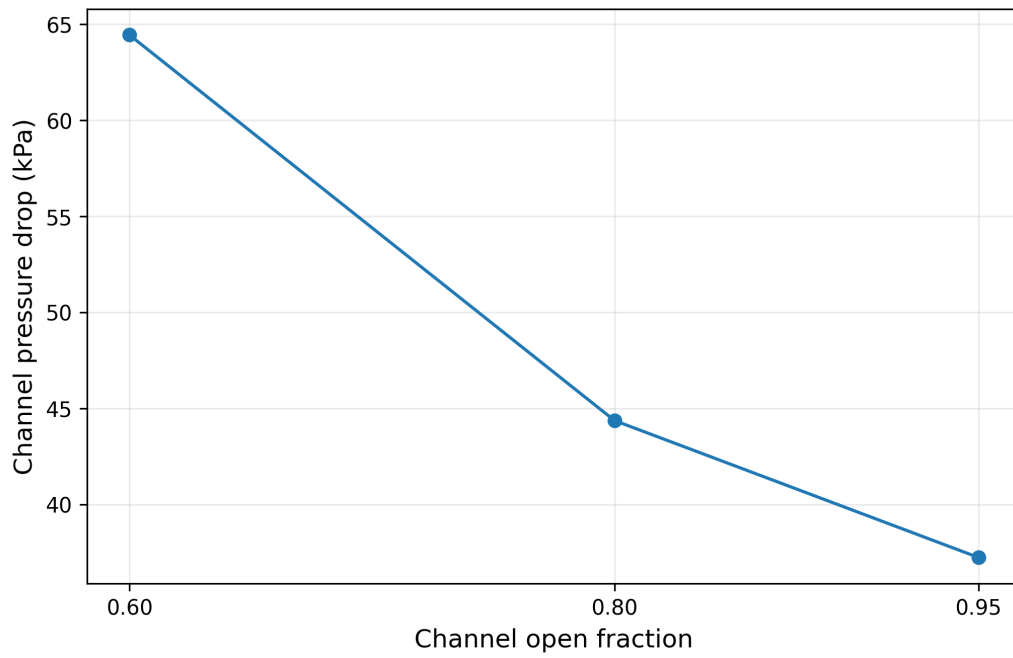


Figure 4.10: Channel pressure drop versus open fraction.

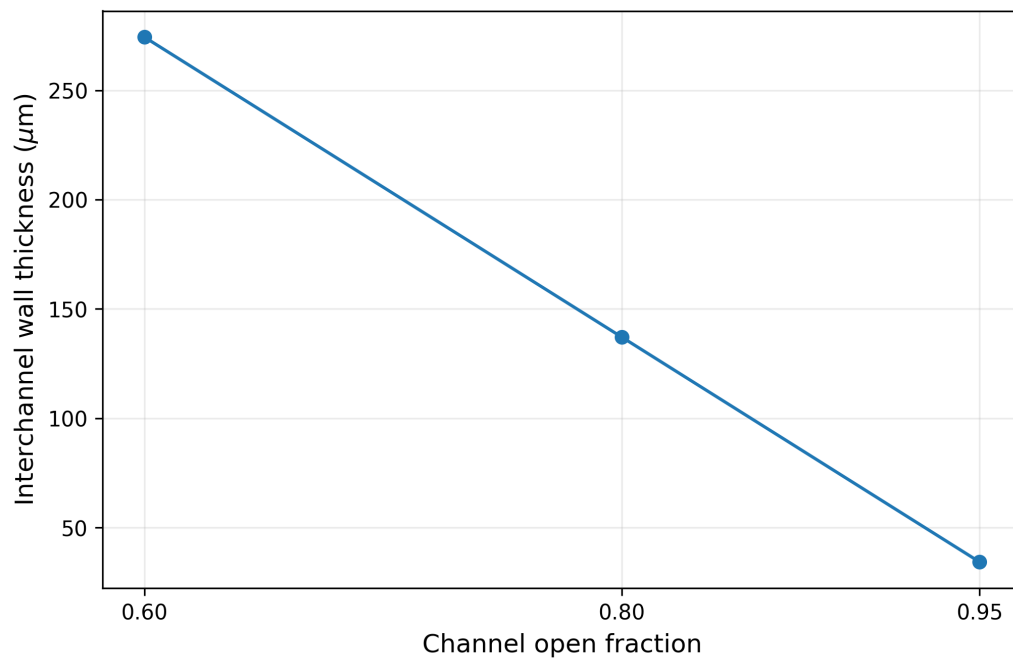


Figure 4.11: Remaining wall thickness versus open fraction.

4.6 Effect of Jet Pitch Ratio and Array Density

For each JPR, the usable jet footprint was populated with as many $300\text{ }\mu\text{m}$ nozzles as the prescribed spacing permitted. Total jet mass flow remained fixed at 4 g/s . Jet count, per-nozzle flow, velocity, and Reynolds number therefore changed simultaneously, making this a system-level JPR/array-density study rather than a pure fixed-nozzle-count pitch experiment.

Maximum temperature decreased from approximately 466 K at JPR 2 to approximately 455 K at JPR 6, then increased slightly to approximately 456 K at JPR 9. JPR 4 produced the smallest chip temperature span, approximately 140 K . Jet total-pressure loss increased from approximately 1.17 kPa at JPR 2 to approximately 126 kPa at JPR 9.

Table 4.5: Results of the variable-count, fixed-total-flow JPR/array-density sweep.

JPR	Array	Flow/jet (g/s)	T_{max} (K)	ΔT_{chip} (K)	Jet ΔP_{total} (kPa)	$Re_{j,est}$
2	7×7	0.082	466	146	1.17	346
4	4×4	0.250	459	140	9.14	1,061
6	3×3	0.444	455	148	25.81	1,886
9	2×2	1.000	456	156	125.98	4,244

Maximum temperature decreased as JPR increased from 2 to 6, after which the improvement largely plateaued. JPR 6 produced the lowest maximum temperature, while JPR 4 produced the smallest chip temperature span. This indicates that thermal performance is controlled by a balance between spatial jet coverage and the momentum available to each individual jet.

Jet pressure loss increased strongly as the array became sparser and the fixed total jet flow was distributed among fewer nozzles. Sparse arrays therefore increased per-nozzle momentum but imposed a substantial hydraulic penalty.

The intermediate optimum results from competing effects of spatial coverage and individual jet momentum. Dense arrays distribute the fixed total flow over many stagnation regions but reduce the momentum available to each jet. Sparse arrays increase per-nozzle

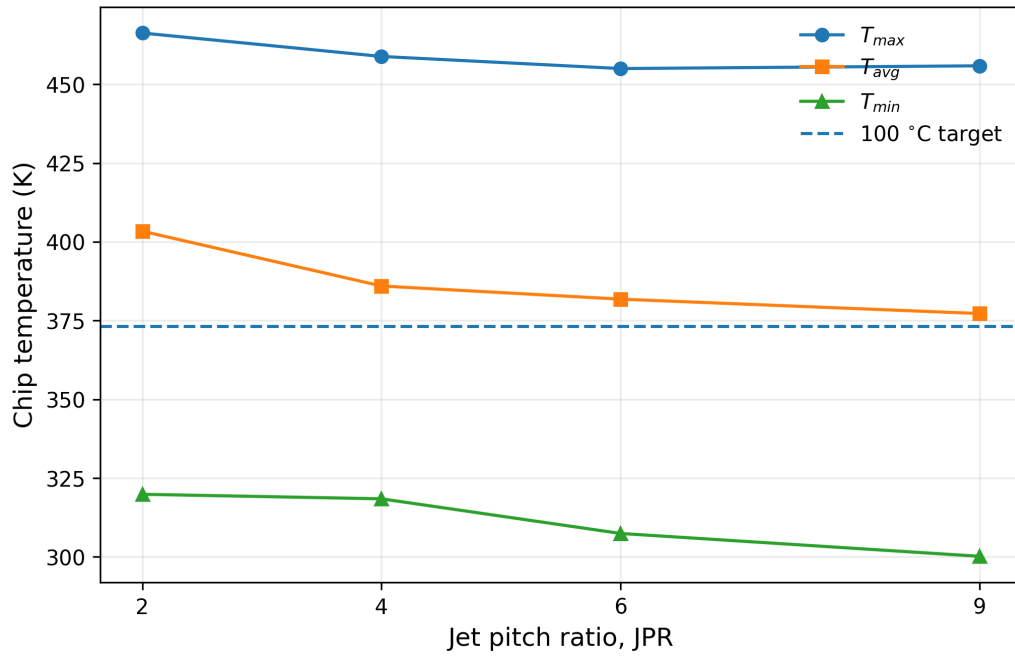


Figure 4.12: Chip temperature metrics for the variable-count, fixed-total-flow JPR/array-density study.

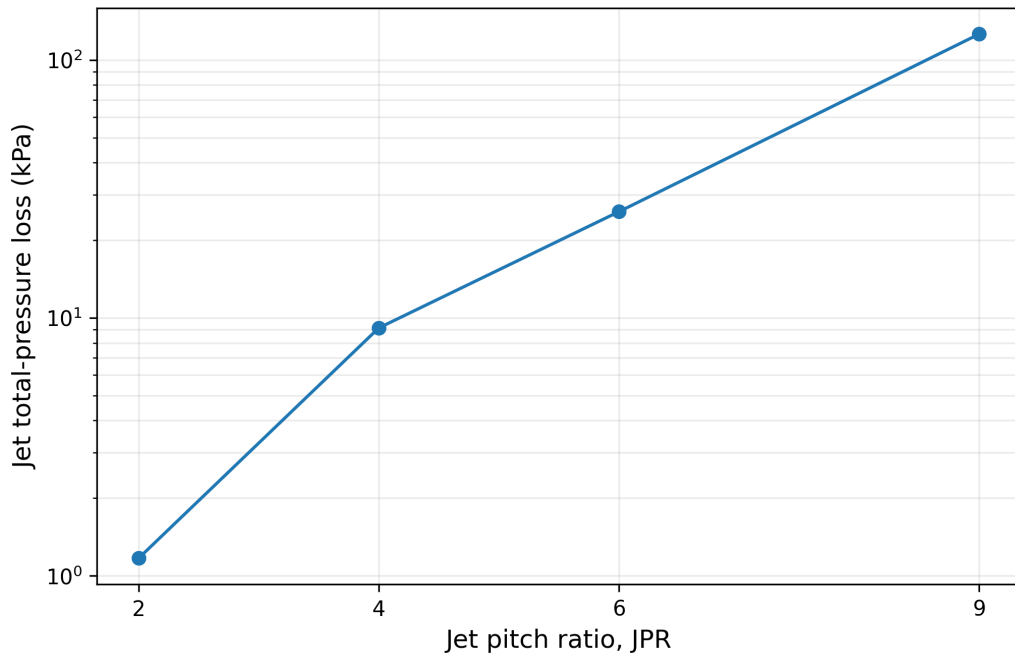


Figure 4.13: Jet total-pressure loss versus jet pitch ratio.

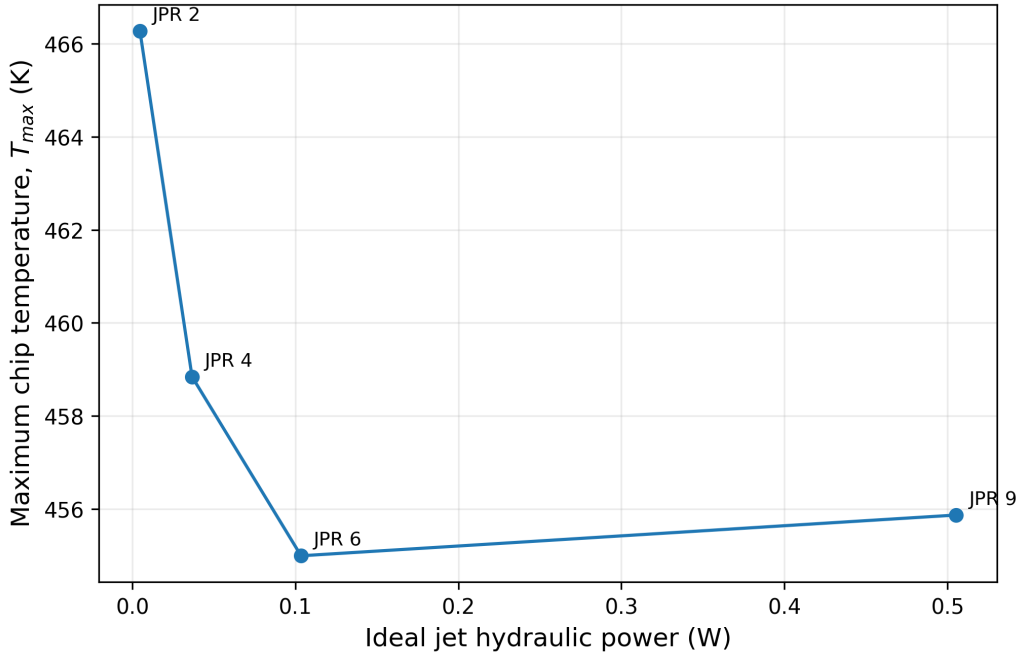


Figure 4.14: Thermal-hydraulic tradeoff for the jet-array study.

momentum but reduce direct spatial coverage and worsen temperature uniformity.

JPR 6 minimized maximum temperature, but improved it by only approximately 4 K relative to JPR 4 while requiring approximately 2.8 times the jet pressure loss and producing a larger chip temperature span. JPR 4 was therefore selected as the preferred thermal-hydraulic compromise. This intermediate range is qualitatively consistent with multiple-jet studies reporting favorable intermediate pitch ratios, although the variable-count formulation used here is not directly equivalent to a fixed-array spacing experiment [9].

4.7 Selected Hybrid Configuration

The selected configuration combines 27 microchannels per microchannel layer, an open fraction of 0.80, a 0.10 mm microchannel edge margin, a total microchannel mass flow rate of 25 g/s, and a 4×4 array of $300 \mu\text{m}$ jets at JPR 4 with a total jet mass flow rate of 4 g/s. The configuration was solved independently after completion of the individual parameter studies.

The model predicted a maximum chip temperature of approximately 374 K (101 °C), an average chip temperature of approximately 339 K (66 °C), a chip temperature span of approximately 77 K, and a maximum chip-to-inlet thermal resistance of approximately 0.115 K/W. The selected resistance is therefore within approximately 1% of the 0.1143 K/W effective resistance required to satisfy the adopted 100 °C design criterion. Relative to the baseline, maximum temperature decreased by approximately 103 K, average temperature by approximately 79 K, temperature span by approximately 79 K, and maximum thermal resistance by approximately 56%.

Table 4.6: Comparison of the baseline and selected hybrid configurations.

Metric	Baseline	Selected configuration	Change
Maximum chip temperature	477 K	374 K	−103 K
Average chip temperature	418 K	339 K	−79 K
Chip temperature span	156 K	77 K	−79 K
Maximum thermal resistance	0.262 K/W	0.115 K/W	−56%
Ideal total hydraulic power	0.328 W	8.38 W	×25.6

The selected maximum remains approximately 1 K above the 100 °C design target and is therefore described as approaching, rather than strictly satisfying, that target. The improvement results from the combined effects identified in the parameter studies: increased channel flow increases coolant heat-removal capacity, increased channel density reduces the characteristic solid-conduction distance, reduced edge margin suppresses perimeter hotspots, the selected open fraction balances coolant area against remaining solid material, and the jet array provides additional localized heat removal.

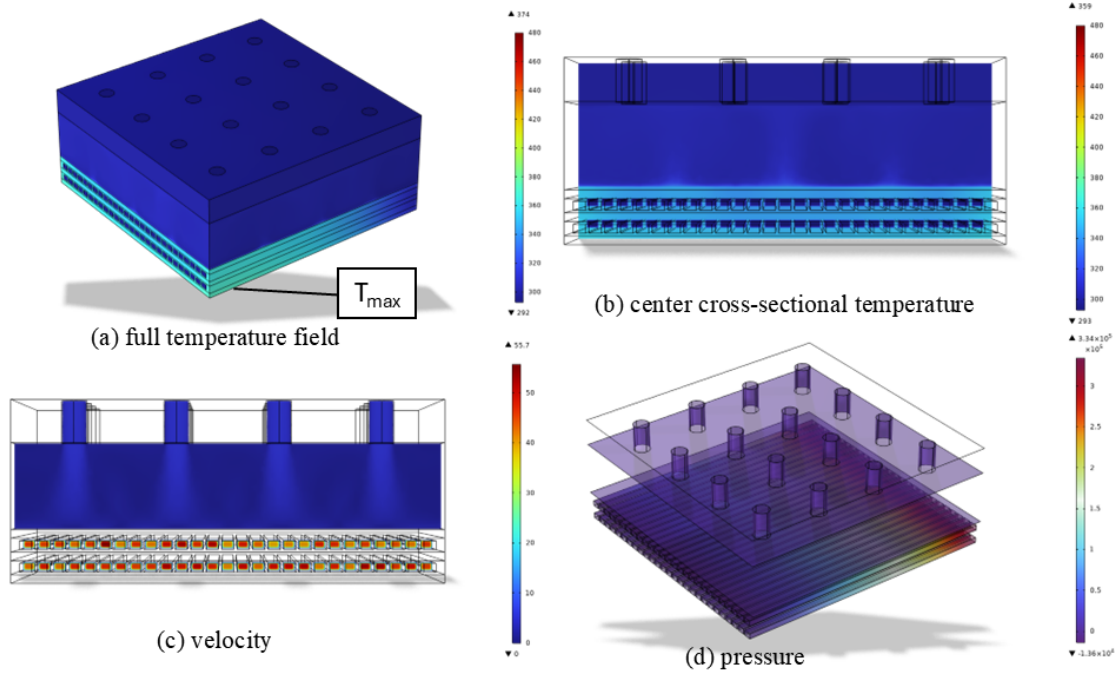


Figure 4.15: Thermal and coolant-flow fields for the selected hybrid configuration: (a) chip temperature with the maximum-temperature location indicated, (b) cross-sectional temperature through the stacked structure, (c) coolant velocity magnitude, and (d) coolant gauge pressure. The temperature panels use the same 293-480 K scale as the baseline fields in Fig. 4.1.

4.8 Selected-Configuration Validation

The selected configuration was subjected to additional conservation, thermodynamic, hydraulic, and reduced-order thermal checks to evaluate whether the predicted performance remained physically and numerically consistent. These checks supplement the parameter-study trends and provide independent or complementary tests of the magnitude and consistency of the final numerical solution.

4.8.1 Mass and Energy Conservation

The selected configuration exhibited an absolute mass imbalance of approximately 0.0008%, indicating close agreement between the prescribed inlet mass flow and calculated outlet flow.

A direct COMSOL total-energy-flux balance was also performed using surface integration of the Heat Transfer interface total-energy-flux variable over the microchannel

and jet inlet and outlet boundaries. The integrated microchannel inlet and outlet energy fluxes were -28.349 W and 600.52 W, respectively, while the corresponding jet inlet and outlet fluxes were -0.0387 W and 128.10 W. Retaining the COMSOL outward-normal sign convention, the net energy gained by the coolant was

$$\dot{Q}_{\text{coolant}} = 600.52 + 128.10 - 28.349 - 0.0387 = 700.23 \text{ W}. \quad (4.1)$$

Relative to the imposed 700 W volumetric heat load, the numerical energy balance recovered approximately 100.03% of the generated heat, corresponding to an absolute closure residual of approximately 0.03% . The close agreement provides strong evidence that the converged selected-configuration solution satisfies global energy conservation.

The branch-level balance also quantifies the relative contribution of the two cooling mechanisms. The microchannel branch gained approximately 572.17 W, corresponding to approximately 81.7% of the total coolant heat gain, while the jet branch gained approximately 128.06 W, or approximately 18.3% . The microchannels therefore provide the majority of the distributed heat removal under the selected operating condition, while the jet branch supplies a smaller but non-negligible parallel heat-removal path.

4.8.2 Single-Phase Coolant Consistency

The maximum water temperature in the selected configuration was approximately 369.34 K (96.2 °C). Because this maximum occurred at a solid-fluid interface, pressure was evaluated immediately within the adjacent water domain at essentially the same location. The calculated local gauge pressure was approximately 7.03 kPa. With the 1 -atm reference pressure used by the COMSOL Laminar Flow formulation, the corresponding absolute pressure was approximately

$$p_{\text{abs}} \approx 101.325 + 7.025 = 108.35 \text{ kPa}. \quad (4.2)$$

The saturation temperature of water at this pressure is approximately 375 K (102 °C), producing a local subcooling margin of approximately

$$\Delta T_{\text{sub}} = T_{\text{sat}} - T_{\text{water,max}} \approx 6 \text{ K.} \quad (4.3)$$

The maximum-temperature coolant region therefore remained below the corresponding local saturation condition. This result supports the single-phase water assumption at the hottest coolant location in the selected configuration. The calculated margin is finite rather than large, however, and phase-change behavior could become relevant if heat load were increased or local pressure were reduced substantially beyond the conditions examined here.

4.8.3 Analytical Microchannel Pressure-Drop Comparison

The selected configuration contains two microchannel layers with 27 channels per layer, corresponding to 54 nominally parallel coolant passages. At a total microchannel mass flow rate of 25 g/s, equal distribution among the channels gives a nominal mean mass flow rate per channel of

$$\dot{m}_{\text{ch}} = \frac{0.025}{54} \approx 4.63 \times 10^{-4} \text{ kg/s.} \quad (4.4)$$

For a channel width of approximately 142.2 μm and height of 100 μm , the hydraulic diameter is approximately 117.4 μm . Using the representative mean-bulk water properties defined in Chapter 3, the nominal mean channel velocity is approximately 32.7 m/s and the Reynolds number is approximately 4,150. This places the selected operating condition above the conventional fully laminar internal-flow range and supports the separate examination of flow-model sensitivity.

Application of the reduced fully developed rectangular-channel relation described in Chapter 3 gives an estimated frictional pressure drop of approximately

$$\Delta P_{\text{analytical}} \approx 324 \text{ kPa} \quad (4.5)$$

over the 5 mm channel length. The selected normal-mesh COMSOL solution predicts an inlet-to-outlet microchannel pressure drop of approximately

$$\Delta P_{\text{COMSOL}} \approx 333 \text{ kPa}. \quad (4.6)$$

The analytical estimate is therefore approximately 3% below the numerical result.

The close numerical agreement supports the physical scale of the COMSOL hydraulic prediction, but it is not interpreted as formal validation of the numerical pressure drop. The reduced relation assumes uniform distribution among the parallel channels, representative constant properties, and a fully developed rectangular-duct flow field, while the three-dimensional numerical model includes temperature-dependent properties and developing and geometric flow effects. In addition, the representative Reynolds number of approximately 4,150 lies beyond the regime in which the fully developed laminar relation is most rigorously applicable. The agreement should therefore be interpreted as a strong reduced-order consistency check rather than as an independent experimental validation.

4.8.4 Representative One-Dimensional Thermal-Resistance Model

An independent reduced-order thermal-resistance model was developed to evaluate whether the characteristic temperature scale predicted by the three-dimensional COMSOL model is consistent with the principal conduction and convection mechanisms of the selected architecture. The model is based on the thermal-ladder framework introduced in Chapter 3 but collapses the distributed network into a representative through-thickness heat-transfer path. It is not intended to reproduce local hotspots, lateral spreading, jet stagnation regions, or the detailed tier-to-tier temperature distribution.

The representative solid path consists of conduction through one 100 μm silicon tier

over the 5×5 mm active footprint. The corresponding conduction resistance is

$$R_{\text{Si}} = \frac{L_{\text{Si}}}{k_{\text{Si}} A_{\text{chip}}}, \quad (4.7)$$

where $L_{\text{Si}} = 100 \mu\text{m}$, $k_{\text{Si}} = 130 \text{ W/(m K)}$, and

$$A_{\text{chip}} = (5 \times 10^{-3})^2 = 2.5 \times 10^{-5} \text{ m}^2. \quad (4.8)$$

The resulting representative silicon resistance is

$$R_{\text{Si}} \approx 0.0308 \text{ K/W}. \quad (4.9)$$

The coolant-side resistance was estimated independently from the selected microchannel geometry and flow conditions. For a channel width of $142.2 \mu\text{m}$ and height of $100 \mu\text{m}$, the hydraulic diameter is

$$D_h = \frac{2w_{\text{ch}}h_{\text{ch}}}{w_{\text{ch}} + h_{\text{ch}}} \approx 117.4 \mu\text{m}. \quad (4.10)$$

Using the representative mean-bulk temperature of approximately 296.5 K established for the reduced-order calculations, the water properties were approximated as $\mu = 9.2 \times 10^{-4} \text{ Pa s}$, $c_p = 4180 \text{ J/(kg K)}$, and $k_f = 0.607 \text{ W/(m K)}$. The corresponding Prandtl number is

$$Pr = \frac{\mu c_p}{k_f} \approx 6.34. \quad (4.11)$$

For the representative Reynolds number of approximately 4,150, the Gnielinski correlation was used as an engineering estimate of the forced-convection heat-transfer scale [16]. The smooth-channel friction factor is

$$f = (0.79 \ln Re - 1.64)^{-2}, \quad (4.12)$$

and the Nusselt number is estimated from

$$Nu = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}. \quad (4.13)$$

For the selected operating condition,

$$f \approx 0.041, \quad Nu \approx 32. \quad (4.14)$$

The corresponding representative microchannel heat-transfer coefficient is

$$h_{mc} = \frac{Nu k_f}{D_h} \approx 1.65 \times 10^5 \text{ W/(m}^2 \text{ K)}. \quad (4.15)$$

The two microchannel layers contain a total of 54 channels. Using the full channel wetted perimeter, the combined wetted area is

$$A_{wet,mc} = N_{ch,total} [2(w_{ch} + h_{ch})L_{ch}] \approx 1.31 \times 10^{-4} \text{ m}^2. \quad (4.16)$$

The corresponding representative convective resistance is therefore

$$R_{conv,mc} = \frac{1}{h_{mc}A_{wet,mc}} \approx 0.0463 \text{ K/W}. \quad (4.17)$$

The reduced one-dimensional resistance is then

$$R_{1D} = R_{Si} + R_{conv,mc} \approx 0.0771 \text{ K/W}. \quad (4.18)$$

To preserve independence from the numerical heat-split result, the full 700 W device heat load was conservatively assigned to the representative microchannel path. At a total microchannel mass flow rate of 25 g/s, the corresponding bulk coolant temperature rise is

$$\Delta T_{bulk} = \frac{Q}{\dot{m}_{mc}c_p} \approx 6.70 \text{ K}. \quad (4.19)$$

Using the mean channel bulk temperature,

$$T_{\text{bulk,mean}} = T_{\text{in}} + \frac{\Delta T_{\text{bulk}}}{2} \approx 297 \text{ K}, \quad (4.20)$$

the representative one-dimensional chip temperature is

$$T_{1\text{D}} = T_{\text{bulk,mean}} + QR_{1\text{D}} \approx 350 \text{ K}. \quad (4.21)$$

The selected COMSOL configuration produced a volume-averaged chip temperature of approximately 339 K. The reduced-order estimate is therefore approximately 11 K higher than the numerical average, corresponding to a difference of approximately 3.4%. This level of agreement is considered reasonable for a deliberately conservative one-dimensional model that assigns the full 700 W load to the microchannel path and neglects the parallel jet-impingement heat-removal mechanism, three-dimensional heat spreading, developing flow, and spatially varying coolant properties. The comparison therefore supports the characteristic thermal scale of the numerical solution without implying that the reduced model reproduces the complete three-dimensional temperature field.

4.8.5 Mesh Sensitivity

The selected configuration was evaluated using coarse, normal, and fine meshes containing approximately 747,000, 852,000, and 2.75 million domain elements, respectively. Maximum chip temperature, chip temperature span, maximum thermal resistance, and microchannel pressure drop were selected as the principal comparison metrics.

Table 4.7: Mesh-sensitivity results for the selected configuration.

Mesh	T_{max} (K)	ΔT_{chip} (K)	$R_{\text{th,max}}$ (K/W)	ΔP_{ch} (kPa)
Coarse	368	72	0.1064	384
Normal	374	77	0.1150	333
Fine	367	64	0.1047	295

The normal-to-fine refinement changed maximum chip temperature by approximately 1.96%, indicating relatively weak mesh sensitivity of the principal peak-temperature metric. Maximum thermal resistance changed by approximately 9.9%, while microchannel pressure drop decreased by approximately 12.7%. Chip temperature span exhibited the greatest sensitivity, changing by approximately 20.8%.

The three quantities did not converge monotonically across the coarse, normal, and fine meshes. In particular, the normal mesh predicted a somewhat higher maximum temperature than either the coarse or fine mesh, while pressure drop decreased progressively with refinement. The study therefore does not demonstrate complete mesh independence of all reported quantities. Instead, it indicates that the principal maximum-temperature prediction is comparatively insensitive to additional refinement, whereas hydraulic loss and spatial temperature-uniformity metrics retain greater discretization sensitivity.

The normal mesh was retained for the primary parametric comparisons because it provided a practical balance between resolution and computational cost while preserving the major thermal trends. Absolute hydraulic values are interpreted with the mesh sensitivity noted above, and the fine-mesh result provides a bounding comparison for the selected configuration.

4.8.6 Flow-Model Sensitivity

The selected configuration has a nominal microchannel Reynolds number of approximately 4,200, placing the highest-flow microchannel condition outside the conventional fully laminar internal-flow regime. The selected JPR 4 jet configuration has a substantially lower estimated jet Reynolds number of approximately 1,060. The flow-model sensitivity study therefore primarily examines the influence of the high microchannel Reynolds number rather than implying fully turbulent behavior throughout the complete coolant domain.

Because all primary parameter studies were performed using the Laminar Flow interface, an exploratory Turbulent Flow, SST calculation was also evaluated on the selected

normal-mesh configuration. The converged laminar solution was used to initialize the SST calculation. The resulting thermal and hydraulic quantities are compared in Table 4.8.

Table 4.8: Exploratory Laminar-SST flow-model sensitivity for the selected configuration.

Metric	Laminar	SST	Difference
Maximum chip temperature	374 K	376 K	+0.70%
Average chip temperature	339 K	350 K	+3.14%
Chip temperature span	77 K	60 K	−21.9%
Maximum thermal resistance	0.1151 K/W	0.1189 K/W	+3.24%
Microchannel pressure drop	333 kPa	249 kPa	−25.2%
Jet total-pressure loss	9.16 kPa	9.38 kPa	+2.43%

The exploratory SST calculation changed the predicted maximum chip temperature by less than 1% and the maximum thermal resistance by approximately 3%. The selected-design thermal conclusion is therefore not strongly affected by the flow-model change in this comparison: both formulations predict a maximum chip temperature near the 100 °C design target.

The larger differences occur in quantities associated with spatial temperature distribution and hydraulic resistance. Relative to the laminar solution, the SST result reduced the predicted chip temperature span by approximately 22% and the microchannel pressure drop by approximately 25%, while the jet total-pressure loss changed by only approximately 2%. These results indicate that the absolute hydraulic-loss and temperature-uniformity predictions are more sensitive to the assumed flow formulation than the maximum chip temperature.

The SST comparison is interpreted cautiously. The reported sensitivity case was obtained during development of the turbulent formulation, and a subsequent attempt to reproduce the solution using a fully rebuilt reciprocal wall-distance initialization and stationary SST solve did not converge within the available computational resources. The SST values are therefore not treated as independently validated turbulent predictions. Instead, they provide a bounded indication of model-form sensitivity and reinforce the decision to interpret the absolute pressure-drop and temperature-uniformity values with greater uncertainty than

the principal maximum-temperature trend.

The Laminar Flow formulation remains the primary model used for all parametric results and design-selection conclusions in this thesis. Maintaining one formulation across the complete design space preserves internal consistency among the parameter sweeps, while the exploratory SST comparison demonstrates that the principal selected-design temperature conclusion is substantially less sensitive to flow-model choice than the hydraulic predictions.

4.9 Thermal-Hydraulic Design Implications

The dominant system-level hydraulic penalty of the selected configuration is associated with the embedded microchannel branch. The inlet-to-outlet microchannel pressure drop is approximately 333 kPa (3.3 bar). At a total microchannel mass flow rate of 25 g/s, corresponding to approximately 1.50 L/min for water, this pressure difference is equivalent to approximately 34 m of water head.

The corresponding ideal microchannel hydraulic power is

$$P_{\text{hyd,mc}} = \Delta P_{\text{mc}} \frac{\dot{m}_{\text{mc}}}{\rho} \approx 8.34 \text{ W}. \quad (4.22)$$

The jet branch operates at a total mass flow rate of 4 g/s and a pressure loss of approximately 9.16 kPa, corresponding to an ideal hydraulic power of approximately 0.037 W. The combined ideal fluid power is therefore approximately

$$P_{\text{hyd,total}} \approx 8.38 \text{ W}. \quad (4.23)$$

This ideal hydraulic power corresponds to approximately 1.2% of the imposed 700 W electronic heat load. However, the required pressure head is a more restrictive practical design consideration than the ideal fluid power alone. A physical cooling loop would require a pump capable of supplying approximately 1.5 L/min through the microchannel branch at a pressure rise of at least approximately 333 kPa, with additional pressure capability required

to overcome losses in external tubing, manifolds, fittings, and the heat exchanger.

Actual pump electrical input would exceed the ideal hydraulic power because of finite pump efficiency and balance-of-system losses. The present study therefore establishes the hydraulic operating requirement of the cooling architecture rather than selecting a specific commercial pump.

The independent rectangular-channel calculation predicted a pressure loss of approximately 324 kPa, compared with approximately 333 kPa from the numerical model. The reduced analytical estimate is therefore approximately 3% below the numerical pressure loss. Although the close agreement supports the physical scale of the predicted hydraulic requirement, the comparison is treated as a reduced-order consistency check rather than formal validation because the analytical relation assumes fully developed flow, representative constant properties, uniform flow distribution, and neglects three-dimensional entrance, exit, and geometric effects.

The resulting thermal-hydraulic tradeoff represents an important system-level finding: under severe heat loading, hydraulic head can become a first-order design constraint even when the ideal mechanical pumping power remains small relative to the electronic heat load.

4.9.1 Design Rules Emerging From the Parametric Studies

Table 4.9: Design rules derived from the investigated parameter ranges.

Design variable	Observed behavior	Selected value
Microchannel mass flow	Temperature and nonuniformity decrease with flow, but thermal improvement diminishes while hydraulic demand rises strongly.	25 g/s
Channel count	Higher density strongly improves cooling but increases pressure loss and reduces interchannel ligament.	27
Microchannel edge margin	Most perimeter-hotspot benefit occurs by 0.25-0.10 mm; further reduction produces little additional improvement.	0.10 mm
Channel open fraction	Increasing from 0.60 to 0.80 produces substantial thermal and hydraulic improvement; 0.95 provides little additional T_{max} benefit while reducing remaining solid material.	0.80
Jet pitch ratio / density	Dense arrays improve spatial coverage but reduce momentum per jet at fixed total flow; sparse arrays increase per-jet momentum but reduce uniformity and increase pressure loss.	JPR 4
Jet flow allocation	Fixed total flow provides a controlled comparison of array density and prevents dense arrays from receiving disproportionately greater total coolant flow.	4 g/s total

4.10 Comparison With Published Literature

The selected maximum chip-to-inlet thermal resistance of approximately 0.115 K/W is similar in magnitude to the approximately 0.11 K/W experimental thermal resistance reported by Xiao et al. for a hybrid microchannel and jet-impingement heat sink [5]. This comparison supports the plausibility of the resistance scale but does not constitute direct validation because the heated area, internal geometry, operating conditions, and overall cooling architecture differ.

The present results also reproduce the broader thermal-hydraulic tradeoff identified throughout the microchannel-cooling literature: increasing coolant-side heat-transfer capability and reducing solid conduction distance can improve thermal performance while simultaneously increasing hydraulic and fabrication penalties [3]. The approximately 333 kPa pressure drop of the selected configuration makes this compromise particularly important at the system level.

The jet-array study similarly demonstrates that maximizing either nozzle density or individual jet momentum does not independently produce the preferred design. The favorable range between JPR 4 and JPR 6 is qualitatively consistent with published multiple-jet trends [9], although direct quantitative comparison is limited because nozzle count varied with pitch in the present fixed-total-flow study.

More broadly, the numerical results support the hybrid-cooling mechanism identified in recent literature: embedded microchannels provide distributed bulk heat removal, while impinging jets provide an additional localized cooling mechanism [5], [10]. The direct energy-flux balance provides quantitative evidence of this division in the selected configuration, with approximately 81.7% of the coolant heat gain occurring in the microchannel branch and approximately 18.3% in the jet branch. The present results further demonstrate that the benefit of combining these mechanisms must be evaluated together with coolant allocation, pressure head, temperature uniformity, and geometric constraints rather than by maximizing

either cooling mechanism independently.

4.11 Validation Summary

Table 4.10 summarizes the principal numerical and physical consistency checks performed on the selected configuration.

Table 4.10: Summary of numerical validation and physical-consistency checks for the selected configuration.

Check	Result	Interpretation
Mass conservation	Absolute imbalance $\approx 0.0008\%$	Inlet and outlet mass flow are numerically consistent.
Direct energy-flux closure	700.23 W recovered from 700 W imposed; residual $\approx 0.03\%$	Strong global energy conservation.
Single-phase consistency	$T_{\text{water,max}} \approx 369 \text{ K}$; local subcooling margin $\approx 6 \text{ K}$	Coolant remains below the local saturation condition at the hottest water location, supporting the single-phase assumption.

Continued on next page

Table 4.10 continued

Check	Result	Interpretation
Channel Reynolds number	Representative analytical estimate $\approx 4.2 \times 10^3$	Using water properties evaluated near the mean microchannel bulk temperature, the selected channel condition lies outside an unambiguously low-Reynolds-number laminar regime. The value is used as a reduced-order flow-regime indicator rather than as a directly solved COMSOL quantity.
Microchannel pressure drop	Analytical ≈ 324 kPa; COMSOL ≈ 333 kPa	Both predict pressure loss on the order of 10^2 kPa. The reduced laminar relation underpredicts the numerical result by approximately 3% and is used as an order-of-magnitude check rather than direct validation.
Representative 1D thermal model	$Nu \approx 32$; $h_{mc} \approx 1.65 \times 10^5$ W/(m ² K); $R_{1D} \approx 0.0771$ K/W; $T_{1D} \approx 350$ K versus $T_{avg,COMSOL} \approx 339$ K	The independently estimated conduction and convection resistance scale reproduces the characteristic numerical average-temperature scale closely. The comparison is not interpreted as a prediction of the localized three-dimensional maximum temperature.

Continued on next page

Table 4.10 continued

Check	Result	Interpretation
Mesh sensitivity	Coarse/normal/fine meshes: 0.747, 0.852, and 2.75 million elements. Normal-to-fine changes: $T_{\max} \approx 1.96\%$, $R_{\text{th},\max} \approx 9.9\%$, $\Delta P_{\text{ch}} \approx 12.7\%$, and $\Delta T_{\text{chip}} \approx 20.8\%$	Peak temperature is comparatively insensitive to further refinement, while hydraulic loss and temperature-uniformity metrics retain greater mesh sensitivity. Complete mesh independence is therefore not claimed.
Flow-model sensitivity	Exploratory SST comparison: $T_{\max}$ changed by $+0.70\%$; $R_{\text{th},\max}$ by $+3.24\%$; ΔP_{ch} by -25.2%	The selected maximum-temperature conclusion is comparatively insensitive to flow-model choice, whereas hydraulic loss and temperature uniformity show greater model-form sensitivity. The SST result is treated as exploratory because a fully rebuilt wall-distance-initialized SST solution was not independently reproduced.

Taken together, the completed checks demonstrate close global mass and energy conservation, a physically plausible hydraulic-loss scale, a positive local subcooling margin, and close agreement between the independent one-dimensional thermal model and the characteristic numerical temperature scale. The mesh study quantifies discretization sensitivity, while the exploratory SST comparison provides an additional indication of flow-model sensitivity at the selected high-Reynolds-number microchannel condition.

Chapter 5: Conclusions

The thermal-management problem addressed by this thesis is inseparable from the packaging trajectory of modern high-performance electronics. AI accelerators, chiplet systems, high-bandwidth-memory integration, and 2.5D/3D stacking are increasing compute and interconnect density while reducing the effectiveness of remote heat-removal paths. These packaging approaches are not a short-lived exception to conventional planar design; they represent a major route by which future systems will continue to scale. As a result, cooling must increasingly move closer to the heat source and become part of package co-design rather than remain an external accessory [1], [2], [6].

This thesis developed a staged computational framework that progressed from reduced-order and preliminary single-level cooling models to a fully three-dimensional conjugate heat-transfer model of a stacked hybrid jet-microchannel architecture. The preliminary models were used to examine the expected behavior and practical limitations of conventional external cooling, jet stagnation cooling, and embedded microchannel flow before these mechanisms were combined within the final stacked configuration. The final parameterized model then evaluated the effects of channel mass flow, channel count, edge margin, open fraction, and jet pitch ratio/array density under an intentionally severe 700 W load distributed through three 5 mm $\times$ 5 mm active tiers.

The parametric studies demonstrate that no single variable controls performance. Microchannel mass flow and channel count were the strongest direct levers for reducing peak temperature, but both introduced increasing hydraulic or geometric penalties. Edge margin controlled a distinct perimeter-hotspot mechanism by changing the lateral conduction distance from the chip perimeter to the cooled channel field. Open fraction produced strong benefit up to approximately 0.80 before additional thermal improvement became small relative to the loss of solid wall material. The jet-array study showed an analogous balance: JPR 6 minimized peak temperature, but JPR 4 provided the better combined result in peak

temperature, temperature uniformity, and pressure loss.

The selected configuration combined 27 microchannels per cooling layer, an open fraction of 0.80, a 0.10 mm edge margin, 25 g/s total microchannel flow, and a 4×4 array of 300 μm jets at JPR 4 supplied by 4 g/s total jet flow. It reduced maximum chip temperature from approximately 477 to 374 K and reduced maximum thermal resistance from 0.262 to 0.115 K/W. The design therefore approached the 100 °C target under the imposed extreme load and reduced maximum thermal resistance by approximately 56% relative to the baseline.

The corresponding microchannel pressure drop of approximately 333 kPa is as important as the temperature result. It demonstrates that once embedded liquid cooling is pushed toward the operating limit of a compact 3D package, the dominant design constraint can shift from insufficient heat-transfer coefficient to hydraulic head, channel geometry, and manufacturability. The selected design should therefore be understood as a high-performance thermal-hydraulic compromise within the investigated design space, not as a globally optimized or balance-of-system-ready cooling loop.

The principal validation and consistency checks support the robustness of the selected thermal result while also identifying the quantities that remain more model-sensitive. In the three-level mesh study, the normal-to-fine change in maximum chip temperature was approximately 1.96%, indicating comparatively weak mesh sensitivity of the peak-temperature prediction, although pressure drop and temperature span remained more sensitive to refinement. An exploratory SST flow-model comparison similarly changed maximum chip temperature by less than 1%, while producing substantially larger changes in channel pressure drop and temperature span, indicating that the principal thermal conclusion is more robust than the absolute hydraulic and uniformity predictions. The independent rectangular-channel calculation predicted a microchannel pressure drop of approximately 324 kPa, compared with approximately 333 kPa from the numerical model, corresponding to a difference of approximately 3%. Direct integration of the COMSOL total-energy flux recovered 700.23 W from the imposed 700 W heat load, giving an energy-closure residual of only approximately

0.03%. Finally, the hottest coolant location remained approximately 6 K below the local saturation temperature, supporting the single-phase water assumption under the selected operating condition. Together, these checks indicate that the principal thermal conclusions are numerically and physically consistent, while hydraulic loss and detailed temperature uniformity retain greater sensitivity to modeling and discretization assumptions.

The broader contribution is a reproducible framework for translating the qualitative advantages of hybrid cooling into quantitative design rules. The results support a shift toward embedded and near-junction cooling for future stacked electronics, while also showing that cooling architecture, package geometry, flow distribution, and pumping requirements must be designed together. This co-design perspective is likely to become more important as 3D integration and high-power AI hardware continue to increase internal heat density and reduce the margin available to conventional package-level thermal management.

Chapter 6: Future Work

The present study establishes the thermal-hydraulic behavior of a parameterized hybrid microchannel and jet-impingement architecture under steady-state operating conditions. The results also identify several directions in which the architecture could be extended beyond the scope of the present work. In particular, future investigations could address realistic coolant distribution, transient and adaptive operation, package-level interface physics, alternative working fluids, expanded geometric optimization, experimental validation, and integration with the external cooling system.

6.1 Manifold and Flow-Distribution Design

A natural extension of the present architecture is the explicit design of the upstream and downstream coolant-distribution systems. The jet-array study prescribed equal flow among disconnected nozzle inlets while holding total jet flow constant. This formulation isolated the influence of array density and per-jet momentum, but intentionally excluded nozzle-to-nozzle maldistribution and upstream plenum behavior. A future model incorporating a common supply plenum and return manifold could resolve these effects directly and determine whether the thermal advantages observed for the idealized jet array persist under realistic flow distribution.

Manifold design is particularly important for the embedded microchannel network. The present results demonstrate that hydraulic head becomes a first-order design constraint at the flow rates required for aggressive cooling. Alternative inlet and outlet arrangements could reduce effective channel flow length, distribute coolant through additional parallel paths, or divide the chip into independently supplied cooling zones. Such architectures could potentially preserve much of the thermal performance of the selected configuration while substantially reducing pressure loss. Future optimization should therefore treat the channels, manifolds, and jet-distribution system as a coupled hydraulic network rather than

as independent components.

6.2 Transient and Adaptive Cooling

The present work considers steady-state heat generation, whereas operating processors exhibit strongly time-dependent and spatially nonuniform power distributions. Future studies could apply transient heat maps representative of realistic computational workloads to determine the response of the hybrid architecture to moving hotspots, rapid power excursions, and spatially concentrated heat generation. Relevant performance metrics would include peak transient temperature, thermal time constant, temperature overshoot, spatial temperature gradients, and the frequency and magnitude of thermal cycling.

The combination of embedded channels and localized jet cooling also provides a potential platform for adaptive thermal management. Independently controlled flow branches, microvalves, variable-pressure jet arrays, or passive temperature-responsive structures could redistribute coolant toward regions experiencing elevated heat flux. Such concepts would extend the present architecture from a fixed cooling network to a spatially responsive thermal-management system. Recent work involving programmable microfluidics and temperature-responsive or hydrogel-assisted cooling provides potential mechanisms for implementing this behavior [4], [8], [11].

6.3 Package Interfaces and Thermomechanical Reliability

The present model intentionally simplifies several package-level features in order to isolate the thermal-fluid behavior of the proposed cooling architecture. A higher-fidelity package model could incorporate thermal contact resistance, bonding layers, underfill, hybrid bonds or microbumps, through-silicon vias, and spatially varying material properties. These features introduce additional thermal resistances and heat-spreading paths that may alter both the magnitude and location of the maximum temperature.

The thermal problem could also be coupled to structural mechanics to examine

thermomechanical reliability. Large temperature gradients across stacked tiers can generate differential thermal expansion and cyclic stresses in bonding layers and interconnects. A coupled thermal-fluid-structural model would allow cooling performance to be evaluated alongside interconnect stress, warpage, fatigue, and package lifetime. Such an analysis would be particularly valuable for determining whether aggressive localized cooling reduces mean package temperature at the expense of increased local thermal gradients or thermomechanical cycling.

6.4 Alternative Working Fluids and Two-Phase Cooling

Water provides a strong single-phase thermal baseline because of its high specific heat and thermal conductivity, but it is not necessarily the preferred coolant for every implementation. Future studies could apply the same geometry and thermal-hydraulic evaluation framework to dielectric single-phase coolants, water-glycol mixtures, or other fluids selected for electrical compatibility, freeze protection, corrosion resistance, or package-material compatibility. Comparisons should consider viscosity, density, thermal conductivity, specific heat, dielectric properties, pumping requirements, material compatibility, environmental impact, and leak-management requirements in addition to temperature reduction.

A more substantial extension would introduce two-phase cooling. The high local heat fluxes considered in this thesis make evaporation and flow boiling potentially attractive because latent heat transfer can provide substantial heat-removal capacity with comparatively small coolant temperature increases. Such a model would require additional treatment of nucleation, phase distribution, pressure-dependent saturation, vapor transport, flow instability, and critical heat-flux limitations. The single-phase architecture developed here could provide a geometric basis for investigating where and under what operating conditions a transition to controlled two-phase cooling becomes advantageous.

6.5 Expanded Geometric and Multi-Objective Optimization

The parameter studies conducted in this work identify several influential design variables but sample only a discrete portion of a much larger design space. Future optimization could simultaneously vary channel width, height, aspect ratio, pitch, edge margin, flow length, nozzle diameter, nozzle shape, standoff ratio, jet spacing, and spatial jet placement. Nonuniform channel or nozzle distributions could also be designed specifically around known hotspot locations rather than using spatially uniform arrays.

Such studies would benefit from a multi-objective formulation rather than optimization of maximum temperature alone. Candidate objective functions could include maximum temperature, temperature nonuniformity, pressure drop, pumping power, coolant flow rate, and package volume. Manufacturability could be incorporated through constraints on minimum wall thickness, channel aspect ratio, lithographic or etching tolerances, sealing requirements, pressure containment, and assembly processes. The resulting Pareto space would provide a more useful design tool than a single temperature-minimizing configuration by explicitly quantifying the cost of additional thermal performance.

6.6 Experimental Demonstration and Validation

Experimental investigation represents an important next step beyond the computational framework developed in this thesis. A representative test article containing embedded microchannels and a jet-cooled surface could be used to measure pressure loss, coolant temperature rise, surface temperature distribution, and overall thermal resistance under controlled heat loads. These measurements would provide direct validation of the conjugate heat-transfer predictions and could quantify effects that are difficult to represent numerically, including surface roughness, manufacturing tolerances, leakage, imperfect bonding, flow maldistribution, and contact resistance.

A staged experimental program could begin with individual microchannel and

jet-impingement test structures before progressing to a stacked hybrid assembly. Such an approach would allow the hydraulic and thermal behavior of each cooling mechanism to be characterized independently before their interaction is examined in the complete architecture. Experimental data could subsequently support refinement of turbulence, conjugate heat-transfer, and package-interface models for increasingly realistic stacked devices.

6.7 Chip-to-System Thermal Co-Design

Finally, the proposed architecture could be extended from a chip-scale thermal-management concept to a complete cooling-system model. The present work demonstrates that high thermal performance can coexist with substantial hydraulic-head requirements, making the interaction between the embedded coolant network and the external loop an important system-level problem. Future models could couple the chip-scale pressure-flow characteristics to pump performance, tubing, valves, manifolds, heat exchangers, and facility cooling infrastructure.

This extension would permit optimization based on total system performance rather than chip temperature alone. Metrics such as pump electrical power, heat-rejection temperature, cooling-system coefficient of performance, coolant inventory, package volume, and facility-level energy consumption could be considered alongside junction temperature and thermal resistance. Ultimately, such an approach would treat the cooling architecture, electronic package, and external thermal infrastructure as a coupled design problem. This system-level perspective represents a logical progression from the hybrid near-junction cooling architecture investigated in the present work toward practical implementation in future high-power 2.5D and 3D electronic systems.

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Appendix A: Computational Model, Data, and Reproducibility Archive

The computational models, exported numerical data, and post-processing materials associated with this thesis are retained as a reproducibility archive. The archive contains the COMSOL Multiphysics models used for the baseline configuration, parameter studies, selected configuration, mesh-sensitivity study, and exploratory flow-model sensitivity calculation, together with the numerical exports used to generate the reported tables and figures.

COMSOL Multiphysics version 6.3.0.420 was used for the final computational studies. Because the native COMSOL model files are large and require compatible licensed software, they are not embedded within the thesis PDF. The archived model and data files are available from the author or thesis advisor subject to George Washington University data-storage policies and availability of a compatible COMSOL installation.

Post-processing of the principal parametric results is consolidated in the Jupyter notebook `thesis_results_plots.ipynb`. The notebook is designed as a portable plotting package in which the original COMSOL CSV exports are stored in the same directory as the notebook. File paths are resolved relative to the notebook working directory, allowing the complete plotting folder to be moved without modifying absolute paths. A single setup cell loads the source data and calculates the limited derived quantities required for plotting, after which each thesis graph is generated in an independent cell. Each figure is displayed inline for inspection and is also exported as a 300-dpi PNG using a stable filename.

The notebook additionally contains a numerical-audit cell that reports the principal rounded values used in the parameter-study tables. This provides a direct check between the source COMSOL exports, plotted results, and values reported in the thesis. Analytical validation calculations described in Chapter 3, including the reduced pressure-drop and thermal-resistance models, are documented independently from the plotting notebook and are not used to modify the underlying COMSOL result data.

The principal archive contents are summarized in Table A.1.

Table A.1: Principal contents of the computational reproducibility archive.

File or group	Purpose
COMSOL model files	Native models for the baseline configuration, parameter sweeps, selected configuration, mesh-sensitivity study, and exploratory flow-model sensitivity calculation
thesis_results_plots.ipynb	Primary reproducible post-processing notebook. Loads the original COMSOL CSV exports, calculates plotting quantities, displays each thesis graph inline, exports stable 300-dpi PNG files, and provides a numerical audit of the plotted parameter-study values
3D_Channel_mdot_sweep.csv	Microchannel mass-flow parameter-study results used for Figs. 4.2-4.4
3D_Channel_N_sweep.csv	Microchannel channel-count parameter-study results used for Figs. 4.5-4.7
3D_Channel_edgemarkin_sweep.csv	Microchannel edge-margin parameter-study results used for the numerical portion of Fig. 4.8
3D_Open_area_sweep.csv	Channel open-fraction parameter-study results used for Figs. 4.9-4.11
3D_JPR_sweep.csv	Fixed-total-flow jet pitch-ratio and array-density results used for Figs. 4.12-4.14
3D_Parametrized_Coarse.csv, 3D_Parametrized_Normal.csv, 3D_Parametrized_Fine.csv 3D_Basic.csv	Coarse-, normal-, and fine-mesh results used for the selected-configuration mesh-sensitivity comparison
3D_Hybrid_final.csv	Numerical export from the baseline three-dimensional hybrid configuration
Notebook-generated PNG files	Numerical export from the selected three-dimensional hybrid configuration
COMSOL image exports and figure source files	Stable figure outputs generated directly by the independent plotting cells, including the mass-flow, channel-count, edge-margin, open-fraction, jet-array, and mesh-sensitivity plots
	Temperature, velocity, pressure, geometry, and mesh visualizations used for the COMSOL-based and composite thesis figures

To reproduce the plotted results, the notebook and its source CSV files are placed in a common directory. The setup and data-loading cell is executed once, after which any individual figure cell may be rerun independently. Figure dimensions and presentation settings can therefore be adjusted without rerunning unrelated plots or modifying the original COMSOL exports.

Appendix B: Additional Figures

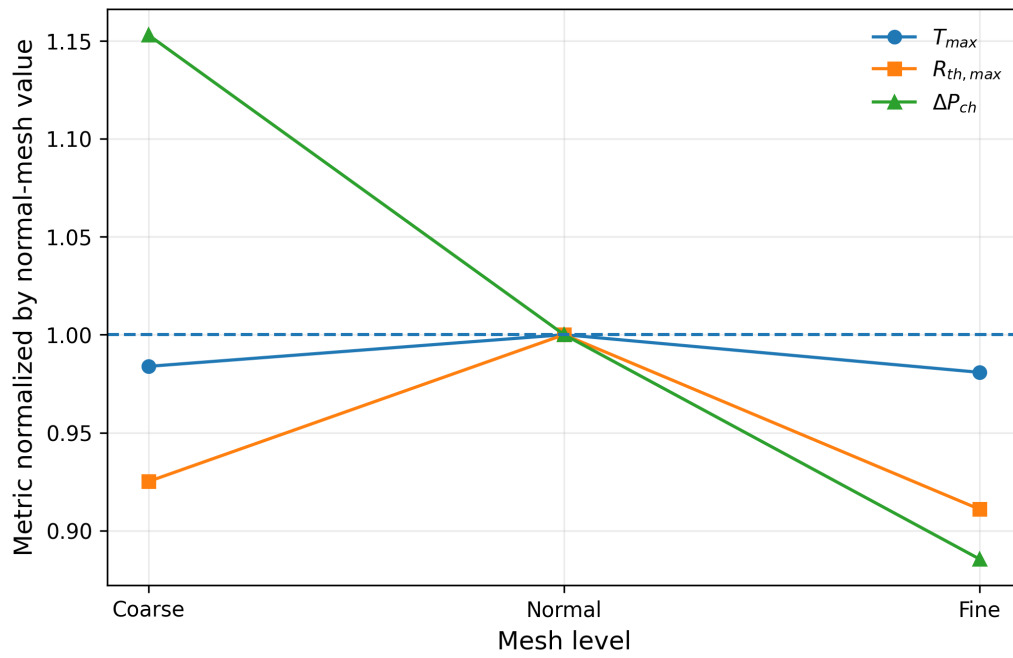


Figure B.1: Mesh-sensitivity results for the selected hybrid configuration.